

Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

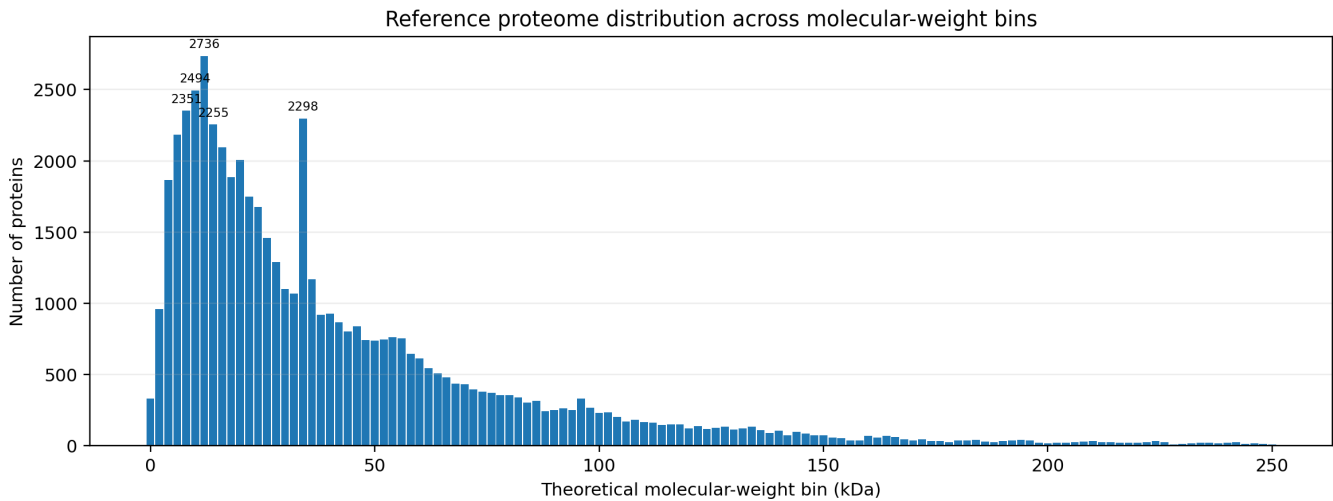
Metric	Value
Proteins parsed	54,857
Molecular-weight bins	294
Bin width	2 kDa
MW range	0.66 to 3925.54 kDa
Median MW	30.14 kDa
Mean MW	47.01 kDa
Proteins <= 50 kDa	38,071 (69.4%)
Proteins 50-100 kDa	11,093 (20.2%)
Proteins > 100 kDa	5,693 (10.4%)
High-priority bins	30
Elevated-priority bins	44
Routine-priority bins	220
Source FASTA	mouse_UP000000589.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

Molecular-weight distribution of the reference proteome.

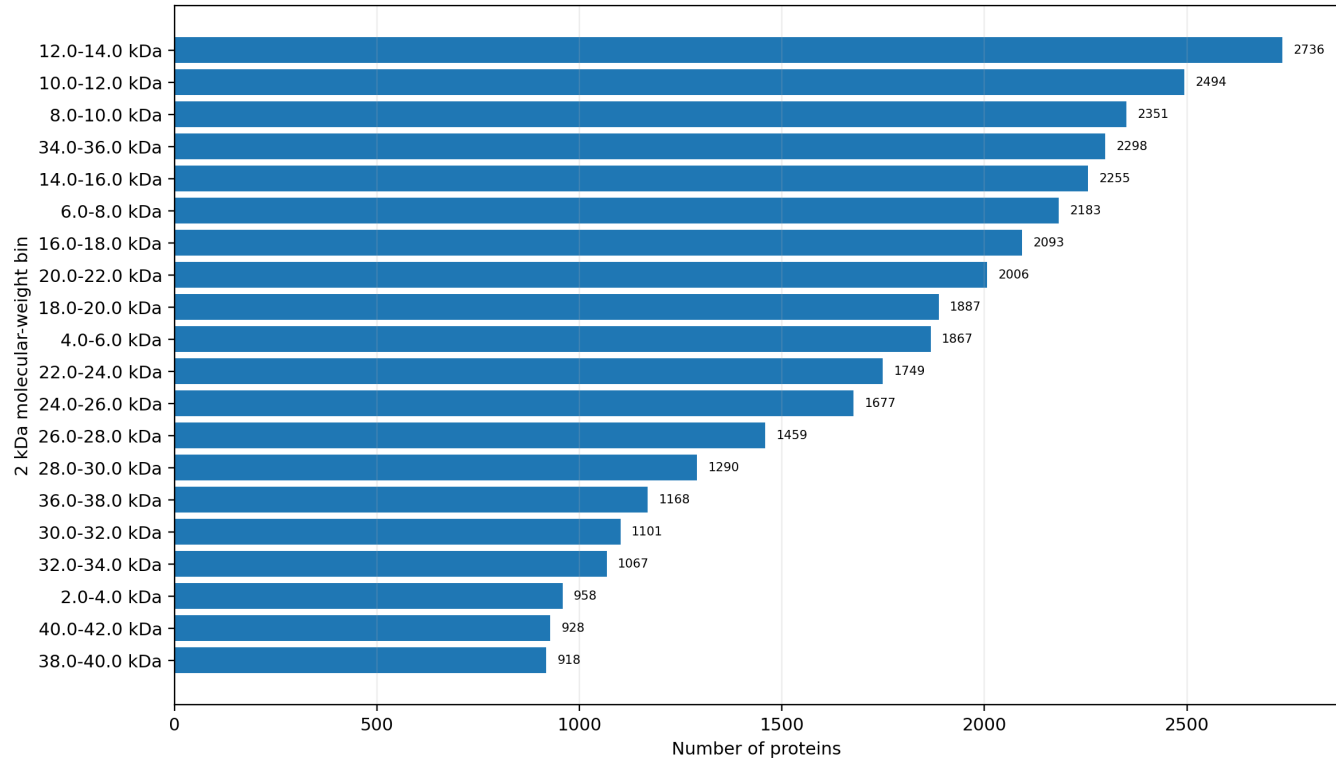
Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.



Most crowded molecular-weight neighborhoods.

The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.

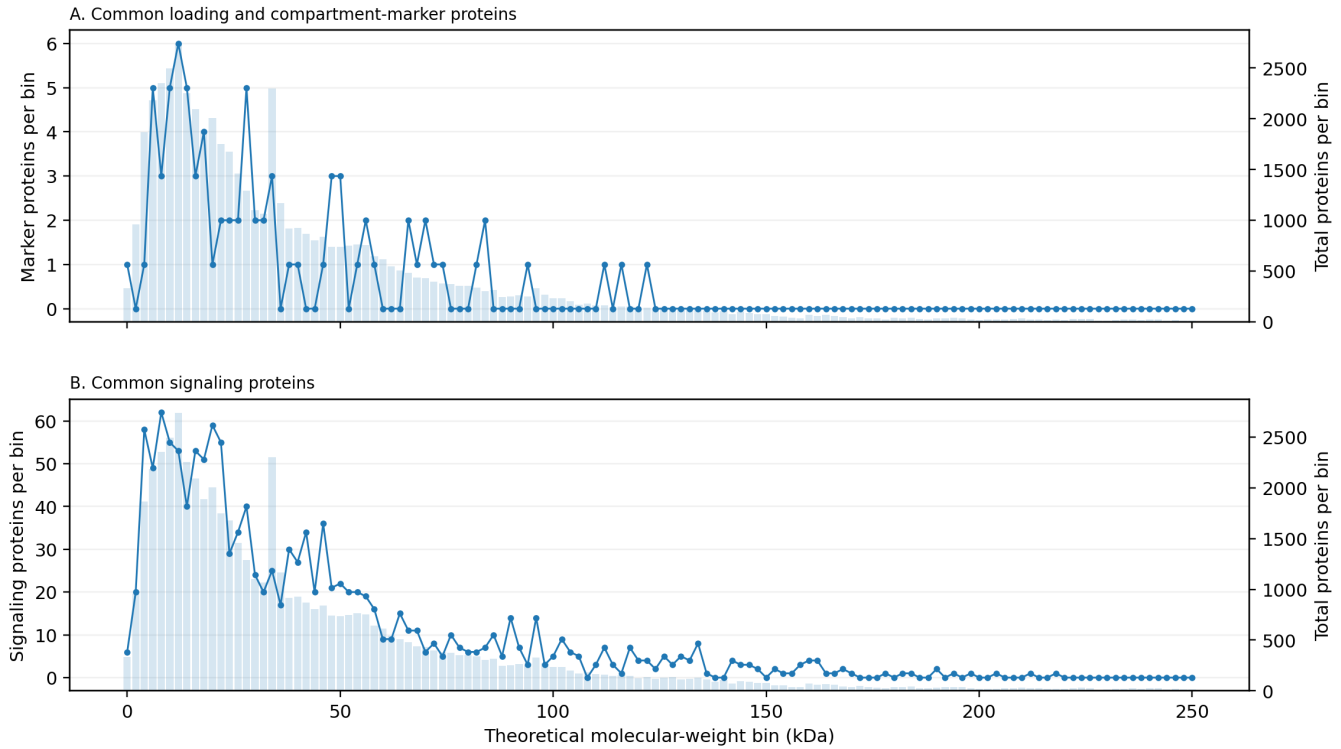
Top 20 most crowded molecular-weight bins



Distribution of common western blot markers and signaling proteins across molecular-weight bins.

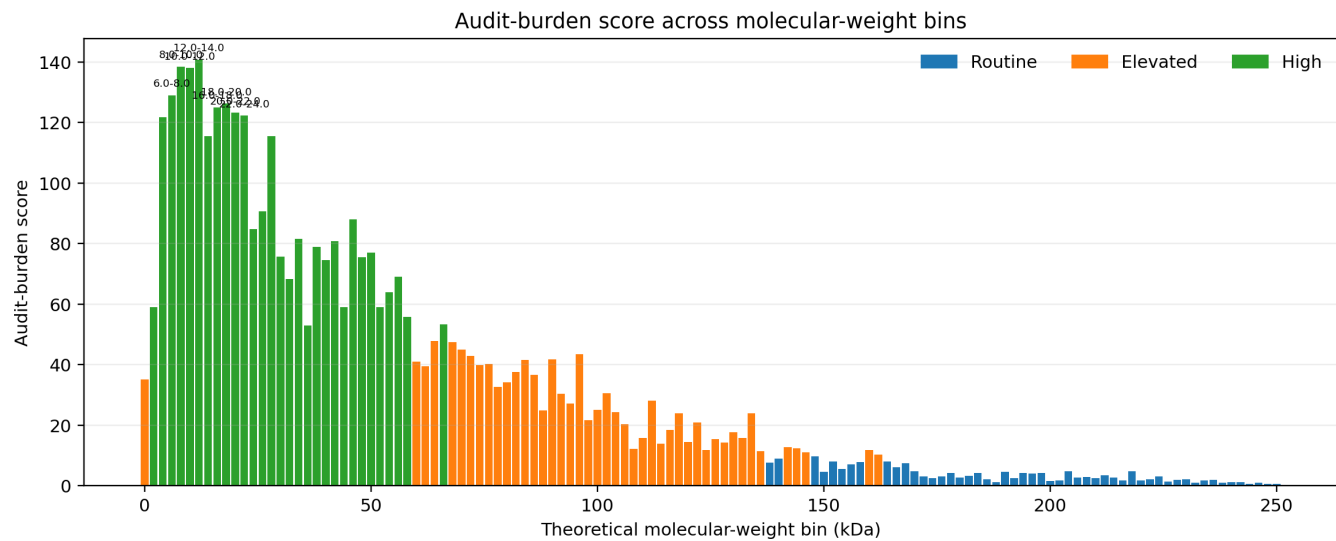
The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.

Audit-relevant annotations across molecular-weight bins



Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
12.0-14.0 kDa	2736	6	53	high	140.8
8.0-10.0 kDa	2351	3	62	high	138.5
10.0-12.0 kDa	2494	5	55	high	138.0
6.0-8.0 kDa	2183	5	49	high	129.0
18.0-20.0 kDa	1887	4	51	high	126.2
16.0-18.0 kDa	2093	3	53	high	125.0
20.0-22.0 kDa	2006	1	59	high	123.2
22.0-24.0 kDa	1749	2	55	high	122.2
4.0-6.0 kDa	1867	1	58	high	121.8
14.0-16.0 kDa	2255	5	40	high	115.5
28.0-30.0 kDa	1290	5	40	high	115.5
26.0-28.0 kDa	1459	2	34	high	90.8
46.0-48.0 kDa	840	1	36	high	88.0
24.0-26.0 kDa	1677	2	29	high	84.8
34.0-36.0 kDa	2298	3	25	high	81.5
42.0-44.0 kDa	869	0	34	high	80.8
38.0-40.0 kDa	918	1	30	high	79.0
50.0-52.0 kDa	738	3	22	high	77.0
30.0-32.0 kDa	1101	2	24	high	75.8
48.0-50.0 kDa	742	3	21	high	75.5
40.0-42.0 kDa	928	1	27	high	74.5
56.0-58.0 kDa	754	2	19	high	69.0
32.0-34.0 kDa	1067	2	20	high	68.2
54.0-56.0 kDa	764	1	20	high	64.0
2.0-4.0 kDa	958	0	20	high	59.0

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
12.0-14.0 kDa	2736	6	53	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	, Atp8a1, Ms4a15, Tgfbtrap1, Nrg2, , Rtca, Pemt [Lamin family], St6galnac6, , , Pum2, Slk19, Zfp655, Pex39, Rbbp5 [Histone family], Slurp1, Tsr3
8.0-10.0 kDa	2351	3	62	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Ubxn2a, Vmn2r83, Gm6993, Slc16a11, Krtap6-1, Styx, Litaaf, Sp140i2, Rcbtb2, Ncl, Cox8c [Mitochondrial markers], Grip1 [Actin family], Rad9a, Asph, D2hgdh, Ppid, Cystm1, A630089N07Rik
10.0-12.0 kDa	2494	5	55	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Evc, Dbi, Ccdc71, A030005L19Rik, Arhgef40, Tmbim1, Clec4n, Kcnj13, Arhgef39, Lce3a, Zfand2a, Gm27235, Ctdnep1 [Mitochondrial markers], Mppe1, Acbd7, Gm17655, Taf11, Sgk1
6.0-8.0 kDa	2183	5	49	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Krtdap, Acbd4, Mlst8, Eri1, Stat5a [JAK-STAT], Retreg1, Xkr5, Shb, S100a14, Plpp5, Gpr12, Slc2a4, Gimap4, Psma3, Ppat, Gm64043, Phf3, Chfr
18.0-20.0 kDa	1887	4	51	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Prdm11, Syp, Ndfip1 [Actin family], Spaca5, Polg2, Ube2l, Zfp426, Pigp, Wtp [Actin family], Jchain, Pcca, Il36a, Ncbp2, Cpne8, Srr, Czib, Sxl1, Utp20
16.0-18.0 kDa	2093	3	53	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Usp15, Slc35b3, Folr1, MNCb-2990, Cep72, Zfp429, Tdrkh, Klf15, Mef2c, Zfp965, Snrpb2, Uqcrc2 [Mitochondrial markers], Rpl32-ps, Grsf1, Dusp3, Bsph1, Pbsn, Arhgap42
20.0-22.0 kDa	2006	1	59	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Bacc1, Smarca2, Sbds, Anxa3, Tmem240, Cd247, Pakap, Qdpr, Obp2b, Csnk2b, Slc47a1, Proser2, Pik3cd, Tm9sf1, Sat1, Septin2, Plin1, Ubr2
22.0-24.0 kDa	1749	2	55	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Il36g, Ctf2, Ppp4r3b, Tll8 [Tubulin family], Ankub1, Arhgef11, Rhoc, Lrrc51, Ctag2l2, Cask, Sh3glb1, Gpank1, Ttc27, Bud13, Ccn2d [Cell-cycle CDK/cyclin], Ube4a, Aq3s2, Hda6c [Histone family]
4.0-6.0 kDa	1867	1	58	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Abhd5, Dnajc1 [Heat shock proteins], Sun2, Cx3cl1, Calm2, Mett1, Sacm1 [Actin family], Rgs1, Cdh5, Sgms1, Chd2, Ddx51, Slc22a21, Sema4c, Fkbp8, Sil1, Galt, Lce1m
14.0-16.0 kDa	2255	5	40	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Creb12, Trbv2-12, Inpp5d, Nop56, Reep6, Tkil1, Ankrd24, Cpne6, Rab38, Gm3676, Cdc163, Nsun3, H2ac21 [Histone family], Cbx6, Sox6, Txncd17, Tsen15, Sp1bn4
28.0-30.0 kDa	1290	5	40	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Krtap4-27, Crybb1, Mff [Mitochondrial markers], Kcnj11, Vps11, Pdyn, Ncor1, , Hoxb9, Cecr2 [Histone family], Pdc, Cdk4 [Cell-cycle CDK/cyclin], Slit2, Coro1b, Fubp3, Rpl8, Pik3ip1 [Actin family], Hdhd3
26.0-28.0 kDa	1459	2	34	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Ubl7, Cdc42bpb [AKT pathway], Mpig6b, Eefsec, Adgrg1, Top3b, Megf11, Mdfi, Lat, Malsu1 [Mitochondrial markers], Clns1a, Tmed4, Naif83, Gm1979, Rrm2, A1597479, Them4, Tmem147
46.0-48.0 kDa	840	1	36	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Sdc3, Serpina1a, Tgfb1i1, Apob, Ccdc71, Aadac12fm2, Mppe1, Ncf1, Cers4, Etv6, Usp30, Nt5dc2, Evi, Slc17a8, Abraxas1, Pdk2 [Mitochondrial markers], Sec1414, Spata6l
24.0-26.0 kDa	1677	2	29	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Metrm, Cyp4b1, Clec2g, Zfand6, Nme5, Rgrip1l, Ropn1, Tecr, Ccs, Ext2, Myo1h, Rims1, Yae1d1, Gm3127, Usl2, Brd8dc, Cbfa212, Pwmp3a
34.0-36.0 kDa	2298	3	25	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Or4b1, Cd276, Slc39a3, Nat1 [Lamin family], Pabpn1, C1qtnf2, Spry1, Tomm40l [Mitochondrial markers], Dhrrs1, Or4c127, Smarca2, Fbh1, Ogn, Pex26, Ori10, Ceacam12, Psmc3, Atad1 [Mitochondrial markers]
42.0-44.0 kDa	869	0	34	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Zranb1, Ppp4r1, Actb12 [Actin family], Marveld3, Acta2 [Actin family], Foxi3, Art3, Opn5, Actc1 [Actin family], Septin1, Syne4, Gna11, Tll5 [Tubulin family], Rnf34, Yy2, Hsd3b3, Bend2, Fut9
38.0-40.0 kDa	918	1	30	Actin family; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	high	Spata6l, Elavl2, Rcn3, Ttc3, Mapk3 [MAPK cascade], Taar8b, Nol4, Pax4, Ncoa4, Cobl1, Borcs6, Cts6, Opcml, Rnf167, Ptgr2, Rnf39, Gmpr2, Wdr45b [Actin family]
50.0-52.0 kDa	738	3	22	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Ash10, Serpina3f, Ccdc91, Runcd3a, Adss2, Zfp113, LOC102638047, Gm38431, Cmpk2 [Mitochondrial markers], Npnt, Cdc25c, Tuba8 [Tubulin family], Ikbbk, Eef1g, Nop58, Zfp36l2, Tom12l, Errf1
30.0-32.0 kDa	1101	2	24	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Svs3b, Snai2, Arhgap36, Eif4a3l1, Ppip5k1, Pdcd2l, Fbxw13, Chid1, Gm9758, Rchy1, Fitm2, Mapre1, 2610028H24Rik, Foxr1, 1700012B07Rik, Ankrd55, Mtmr11, Mmp7
48.0-50.0 kDa	742	3	21	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Mapk10 [MAPK cascade], Smoc1, Dph1, Chst9, Prg4, Lipc, Cse1l, Tead4, Sbf1, Rnf150, Ccdc83, Kynu, Amdhd2, Kcnj16, Entr1, U2surp, Krt23, Dok3
40.0-42.0 kDa	928	1	27	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Mbd4, Tead1, Tada2b, Ptgdr, Sstr5, Lhx2, Hsd17b7, Agbl4, Sp6, Plek2, Pofut1, Lgals9, Nans, Ili10rb, Cxcr1, Lgals9, Gpr27, Rpf1
56.0-58.0 kDa	754	2	19	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; p53 pathway	high	Usp14, Creb5, Irf5, Klc3, Btk, Cyp2c70, Cyp2c66, LOC620551, Prame14, Traf2, Gcm2, Slc18a1, Slc43a3, Antrx1, Nedd4, Rnf217, Cyp214, Gcm2
32.0-34.0 kDa	1067	2	20	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Zmat3, Jmjd6, Gjb1, C1qtnf1, Oser1, Dgkb, Nkx1-2, Ttpa, Fcho2, D17h6553e, Sh2d6, Usf1, Nup85, Btbd19, Eci1 [Mitochondrial markers], Dnase1, Mrgprg, Ccser2
54.0-56.0 kDa	764	1	20	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Zfp810, Tmem200a, Etv4, Nab1, Ralgps2, Tmem161a, Pnliprp2, F10, Ltv1, Vmn2r3, Polr1e, Mapk10 [MAPK cascade], Vmn2r3, Pnma3, Krt24, Slc7a11, Scarb2, Fam114a2
2.0-4.0 kDa	958	0	20	AKT pathway; Actin family; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Traj9, Traj25, Slac3, Akap7, Traj24, Traj31, Hras, Gltd81, Man2a2, Traj41, Mier1, Trgj2, Traj34, Chd7, Aasdh, Sgms2, Traj33, Traj4
44.0-46.0 kDa	803	0	20	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Foxh1, Mtg2 [Mitochondrial markers], Mterf1b [Mitochondrial markers], Serbp1, Vsig1, Rptor, Amt [Mitochondrial markers], Ankn1, Lhx3, Fnta, Clasp2, Npy1r, Vsig1, Mapk8 [MAPK cascade], Cipc [Actin family], Wnk1, Mett14, Gm20509
52.0-54.0 kDa	748	0	20	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Chia, Serpinc1, Ferz1, Immt, Cpsf7, Retreg2, Htr7, Zfp874a, Gsdma3, Krtap16-1, Rmdn3, Fam161a, Agt, Ntrk2, Tekt4, Wrap73, Atg4c, Snx8
58.0-60.0 kDa	645	1	16	Actin family; Cell-cycle CDK/cyclin; ER markers; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Ccl6a, Ccnl2 [Cell-cycle CDK/cyclin], Flo, Clca3a1, Cyp19a1, Eefsec, Espn, Siglecf, Cyp3a59, Btrc, Slc7a1, Igf2bp2, Crnn, Agfg1, Gm10778, Lao1, Klni33, Cct4

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
66.0-68.0 kDa	482	2	11	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	high	Gm8113, Bank1, Zfp345, Zbtb22, Zbtb44, Dcaf8, Fbxw7, Frs1, Gga2, Ryk, Plk5, Lrln3, Btbd9, Hrob, Syk, Frs1, Plcd4, Vmn2r105
36.0-38.0 kDa	1168	0	17	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Vmn1r193, V1ra11, Ankrd1, Ica1, Or2aj6, Or5al5, Frzb, Or9k7, Prss32, Vmn1r84, Spdya, Kcmf1, Tas2r122, Or4a27, Vmn1r227, Septin12, Ilvbl, Or14j6

Common loading/compartiment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
1.37	Tfrc	D3YZB9	D3YZB9_MOUSE	Transferrin receptor (Fragment)	0.0-2.0 kDa
5.60	H3f3a	E0CYN1	E0CYN1_MOUSE	H3.3 histone A (Fragment)	4.0-6.0 kDa
6.08	H3f3a	E0CYR7	E0CYR7_MOUSE	H3.3 histone A (Fragment)	6.0-8.0 kDa
6.22	Gapdh	S4R1W8	S4R1W8_MOUSE	glyceraldehyde-3-phosphate dehydrogenase (phosphorylating) (Fragment)	6.0-8.0 kDa
6.32	Gapdh	S4R1N5	S4R1N5_MOUSE	glyceraldehyde-3-phosphate dehydrogenase (phosphorylating) (Fragment)	6.0-8.0 kDa
7.03	Cox4i1	M0QWX2	M0QWX2_MOUSE	Cytochrome c oxidase subunit 4i1	6.0-8.0 kDa
7.94	Vdac2	D3YUN8	D3YUN8_MOUSE	Non-selective voltage-gated ion channel VDAC2 (Fragment)	6.0-8.0 kDa
8.28	Tbp	D3Z453	D3Z453_MOUSE	TATA box binding protein (Fragment)	8.0-10.0 kDa
8.72	Hsp90ab1	D3Z1R1	D3Z1R1_MOUSE	Heat shock protein 90 alpha (cytosolic), class B member 1 (Fragment)	8.0-10.0 kDa
9.21	Atp5f1a	D6RJ16	D6RJ16_MOUSE	ATP synthase F1 subunit alpha	8.0-10.0 kDa
10.85	Cox4i1	M0QWX7	M0QWX7_MOUSE	Cytochrome c oxidase subunit 4 (Fragment)	10.0-12.0 kDa
11.18	Hsp90aa1	A2A6A2	A2A6A2_MOUSE	Heat shock protein 90, alpha (cytosolic), class A member 1 (Fragment)	10.0-12.0 kDa
11.40	Actb	E9Q2D1	E9Q2D1_MOUSE	Actin, beta (Fragment)	10.0-12.0 kDa
11.43	Tomm20	A0A1D5RLZ6	A0A1D5RLZ6_MOUSE	Translocase of outer mitochondrial membrane 20	10.0-12.0 kDa
11.71	Hsp90aa1	B7ZC49	B7ZC49_MOUSE	Heat shock protein 90, alpha (cytosolic), class A member 1 (Fragment)	10.0-12.0 kDa
12.81	Hspa8	D3Z5E2	D3Z5E2_MOUSE	Heat shock protein 8 (Fragment)	12.0-14.0 kDa
12.96	Hsp90ab1	E9PX27	E9PX27_MOUSE	Heat shock protein 90 alpha (cytosolic), class B member 1	12.0-14.0 kDa
13.30	Lmna	D3YUF7	D3YUF7_MOUSE	Lamin A (Fragment)	12.0-14.0 kDa
13.32	H3f3a	E0CZ27	E0CZ27_MOUSE	Histone H3 (Fragment)	12.0-14.0 kDa
13.66	Tubb3	A0A1D5RM76	A0A1D5RM76_MOUSE	Tubulin, beta 3 class III	12.0-14.0 kDa
13.67	Tbp	E9Q6D9	E9Q6D9_MOUSE	TATA box binding protein (Fragment)	12.0-14.0 kDa
14.06	H2ac1	L7MTW0	L7MTW0_MOUSE	H2A clustered histone 1	14.0-16.0 kDa
15.20	H3f3a	F8WI35	F8WI35_MOUSE	Histone H3	14.0-16.0 kDa
15.39	H3c2	P84228	H32_MOUSE	Histone H3.2	14.0-16.0 kDa
15.40	H3c1	P68433	H31_MOUSE	Histone H3.1	14.0-16.0 kDa
15.65	Tbp	E9Q9Z7	E9Q9Z7_MOUSE	TATA box binding protein (Fragment)	14.0-16.0 kDa
16.06	Rplp0	D3YVM5	D3YVM5_MOUSE	Large ribosomal subunit protein uL10 (Fragment)	16.0-18.0 kDa
16.28	Tomm20	Q9DCC8	TOM20_MOUSE	Mitochondrial import receptor subunit TOM20 homolog	16.0-18.0 kDa
16.66	Actb	E9Q606	E9Q606_MOUSE	Actin, beta (Fragment)	16.0-18.0 kDa
18.45	P4hb	E9Q8G8	E9Q8G8_MOUSE	protein disulfide-isomerase	18.0-20.0 kDa
18.91	Hsp90ab1	E9Q0C3	E9Q0C3_MOUSE	Heat shock protein 90 alpha (cytosolic), class B member 1 (Fragment)	18.0-20.0 kDa
19.36	Tbp	E9Q084	E9Q084_MOUSE	TATA box binding protein (Fragment)	18.0-20.0 kDa
19.53	Cox4i1	P19783	COX41_MOUSE	Cytochrome c oxidase subunit 4 isoform 1, mitochondrial	18.0-20.0 kDa
21.89	Vdac2	D3YZT5	D3YZT5_MOUSE	Non-selective voltage-gated ion channel VDAC2 (Fragment)	20.0-22.0 kDa
22.39	Hsp90aa1	B7ZC50	B7ZC50_MOUSE	Heat shock protein 90, alpha (cytosolic), class A member 1 (Fragment)	22.0-24.0 kDa
22.48	Hsp90ab1	E9Q3D6	E9Q3D6_MOUSE	Heat shock protein 90 alpha (cytosolic), class B member 1 (Fragment)	22.0-24.0 kDa
24.08	Rplp0	S4R1N1	S4R1N1_MOUSE	Large ribosomal subunit protein uL10 (Fragment)	24.0-26.0 kDa
25.13	Sdha	A0A1Y7VJ55	A0A1Y7VJ55_MOUSE	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial (Fragment)	24.0-26.0 kDa
26.74	Vdac2	A0A286YCR8	A0A286YCR8_MOUSE	Non-selective voltage-gated ion channel VDAC2 (Fragment)	26.0-28.0 kDa
27.10	Tbp	F8WJ67	F8WJ67_MOUSE	TATA box binding protein (Fragment)	26.0-28.0 kDa
28.16	Vdac1	F2Z471	F2Z471_MOUSE	Non-selective voltage-gated ion channel VDAC1	28.0-30.0 kDa
28.30	Tbp	F7CLM2	F7CLM2_MOUSE	TATA box binding protein	28.0-30.0 kDa
28.78	Pcna	P17918	PCNA_MOUSE	DNA sliding clamp PCNA	28.0-30.0 kDa
29.44	Actb	E9Q5F4	E9Q5F4_MOUSE	Actin, beta (Fragment)	28.0-30.0 kDa
29.94	Gapdh	S4R257	S4R257_MOUSE	Glyceraldehyde-3-phosphate dehydrogenase (Fragment)	28.0-30.0 kDa
30.45	Vdac2	G3UX26	G3UX26_MOUSE	Non-selective voltage-gated ion channel VDAC2	30.0-32.0 kDa
31.73	Vdac2	Q60930	VDAC2_MOUSE	Non-selective voltage-gated ion channel VDAC2	30.0-32.0 kDa
32.35	Vdac1	Q60932	VDAC1_MOUSE	Non-selective voltage-gated ion channel VDAC1	32.0-34.0 kDa
32.56	Actb	E9Q1F2	E9Q1F2_MOUSE	Actin, beta	32.0-34.0 kDa
34.22	Rplp0	P14869	RLA0_MOUSE	Large ribosomal subunit protein uL10	34.0-36.0 kDa
34.71	Tbp	P29037	TBP_MOUSE	TATA-box-binding protein	34.0-36.0 kDa
35.81	Gapdh	P16858	G3P_MOUSE	Glyceraldehyde-3-phosphate dehydrogenase	34.0-36.0 kDa
38.65	Gapdh	A0A0A0MQF6	A0A0A0MQF6_MOUSE	Glyceraldehyde-3-phosphate dehydrogenase	38.0-40.0 kDa
41.74	Actb	P60710	ACTB_MOUSE	Actin, cytoplasmic 1	40.0-42.0 kDa
47.99	Calr	P14211	CALR_MOUSE	Calreticulin	46.0-48.0 kDa
49.91	Tubb2a	Q7TMM9	TBB2A_MOUSE	Tubulin beta-2A chain	48.0-50.0 kDa
49.91	Tuba1c	P68373	TBA1C_MOUSE	Tubulin alpha-1C chain	48.0-50.0 kDa
49.95	Tubb2b	Q9CWF2	TBB2B_MOUSE	Tubulin beta-2B chain	48.0-50.0 kDa
50.14	Tuba1a	P68369	TBA1A_MOUSE	Tubulin alpha-1A chain	50.0-52.0 kDa
50.15	Tuba1b	P05213	TBA1B_MOUSE	Tubulin alpha-1B chain	50.0-52.0 kDa
50.42	Tubb3	Q9ERD7	TBB3_MOUSE	Tubulin beta-3 chain	50.0-52.0 kDa
54.59	Atp5f1a	D3Z6F5	D3Z6F5_MOUSE	ATP synthase subunit alpha	54.0-56.0 kDa
57.06	P4hb	P09103	PDIA1_MOUSE	Protein disulfide-isomerase	56.0-58.0 kDa
57.34	Tfrc	F7B1E2	F7B1E2_MOUSE	Transferrin receptor	56.0-58.0 kDa
59.75	Atp5f1a	Q03265	ATPA_MOUSE	ATP synthase F(1) complex subunit alpha, mitochondrial	58.0-60.0 kDa
66.79	Lmnb1	P14733	LMNB1_MOUSE	Lamin-B1	66.0-68.0 kDa
67.28	Canx	P35564	CALX_MOUSE	Calnexin	66.0-68.0 kDa
68.78	Hspa8	F7AZ60	F7AZ60_MOUSE	Heat shock protein 8	68.0-70.0 kDa
70.08	Hspa1a	Q61696	HS71A_MOUSE	Heat shock 70 kDa protein 1A	70.0-72.0 kDa
70.87	Hspa8	P63017	HSP7C_MOUSE	Heat shock cognate 71 kDa protein	70.0-72.0 kDa
72.59	Sdha	Q8K2B3	SDHA_MOUSE	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
74.24	Lmna	P48678	LMNA_MOUSE	Prelamin-A/C	74.0-76.0 kDa
83.28	Hsp90ab1	P11499	HS90B_MOUSE	Heat shock protein HSP 90-beta	82.0-84.0 kDa
84.79	Hsp90aa1	P07901	HS90A_MOUSE	Heat shock protein HSP 90-alpha	84.0-86.0 kDa
85.73	Tfrc	Q62351	TFR1_MOUSE	Transferrin receptor protein 1	84.0-86.0 kDa

MW kDa	Gene	Accession	Entry	Protein	MW bin
95.31	Eef2	P58252	EF2_MOUSE	Elongation factor 2	94.0-96.0 kDa
112.98	Atp1a1	Q8VDN2	AT1A1_MOUSE	Sodium/potassium-transporting ATPase subunit alpha-1	112.0-114.0 kDa
116.72	Vcl	Q64727	VINC_MOUSE	Vinculin	116.0-118.0 kDa
123.86	Vcl	A0A286YDJ4	A0A286YDJ4_MOUSE	Vinculin	122.0-124.0 kDa

Complete molecular-weight bin summary, low to high

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
0.0-2.0 kDa	332	1	6	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Mitochondrial markers; p53 pathway	elevated	Gm45021, , Aars1, Ighd1-1, Tsc2, , Tir3, Tor1aip2 [Actin family], Dtx2, Slc2a4, Pou2f1, Ighd1-3, Rnf44, Dab2, Pgppep1, Nrip1 [Actin family], Ttl1,
2.0-4.0 kDa	958	0	20	AKT pathway; Actin family; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Traj9, Traj25, Stac3, Akap7, Traj24, Traj31, Hras, Gli8d1, Man2a2, Traj41, Mier1, Trgj2, Traj34, Chd7, Aasdh, Sgms2, Traj33, Traj4
4.0-6.0 kDa	1867	1	58	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Abhd5, Dnajc1 [Heat shock proteins], Sun2, Cx3cl1, Calm2, Mett11, Sacm11 [Actin family], Rgs1, Cdh5, Sgms1, Chd2, Ddx51, Slc22a21, Sema4c, Fkbp8, Sll1, Gall, Lce1m
6.0-8.0 kDa	2183	5	49	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Krtdap, Acbd4, Mist8, Eri1, Stat5a [JAK-STAT], Retreg1, Xkr5, Shb, S100a14, Plpp5, Gpr12, Slc2a4, Gimap4, Psma3, Ppat, Gm64043, Phf3, Chfr
8.0-10.0 kDa	2351	3	62	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Ubxn2a, Vmn2r83, Gm6993, Slc16a11, Krtap6-1, Styx, Litafd, Sp140I2, Rcbtb2, Ncl, Cox8c [Mitochondrial markers], Grip1 [Actin family], Rad9a, Asph, D2ghdh, Ppid, Cystm1, A630089N07Rik
10.0-12.0 kDa	2494	5	55	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Evc, Dbi, Ccdc71, A030005L19Rik, Arhge40, Tmbim1, Clec4n, Kcnj13, Arhgef39, Lce3a, Zlfand2a, Gm27235, Ctdnep1 [Mitochondrial markers], Mppe1, Acbd7, Gm17655, Taf11, Sgk1
12.0-14.0 kDa	2736	6	53	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	, Atp8a1, Ms4a15, Tgfbtrap1, Nrg2, , Rtca, Pemt [Lamin family], St6galnac6, , , Pum2, Stk19, Zfp655, Pex39, Rbbp5 [Histone family], Slurp1, Tsr3
14.0-16.0 kDa	2255	5	40	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Creb12, Trbv12-2, Inpp5d, Nop56, Reep6, Tkt1, Ankrd24, Cpne6, Rab38, Gm3676, Ccdc163, Nsun3, H2ac21 [Histone family], Cbx6, Sox6, Txndc17, Tsen15, Sptbn4
16.0-18.0 kDa	2093	3	53	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Usp15, Slc35b3, Fcrl1, MNCB-2990, Cep72, Zfp429, Tdrkh, Klf15, Mef2c, Zfp965, Snrpb2, Uqcrc2 [Mitochondrial markers], Rpl32-ps, Grsf1, Dusp3, BspH1, Pbsn, Arhgap42
18.0-20.0 kDa	1887	4	51	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Prdm11, Syp, Ndfip1 [Actin family], Spaca5, Polg2, Ube2i, Zfp426, Pigp, Wip [Actin family], Jchain, Pcca, Il36a, Ncbp2, Cpne8, Srr, Czib, Slx1, Utp20
20.0-22.0 kDa	2006	1	59	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Bacc1, Smarca2, Sbds, Anxa3, Tmem240, Cd247, Pakap, Qdpr, Obp2b, Csnk2b, Slc47a1, Proser2, Pik3cd, Tm9sf1, Sat1, Septin2, Plin1, Ubr2
22.0-24.0 kDa	1749	2	55	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Il36g, Chf2, Ppp4r3b, Ttl8 [Tubulin family], Ankub1, Arhgef11, Rhoc, Lrrc51, Ctag2l2, Cask, Sh3glb1, Gpank1, Tlc27, Bud13, Cnd2 [Cell-cycle CDK/cyclin], Ube4a, Ap3s2, Hdac6 [Histone family]
24.0-26.0 kDa	1677	2	29	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Metn, Cyp4b1, Clec29, Zlfand6, Nme5, Rpgrip11, Ropn1, Tecr, Ccs, Ext2, Myo1h, Rims1, Yae1d1, Gm3127, Usf2, Brd8dc, Cbfa2l2, Pwpp3a
26.0-28.0 kDa	1459	2	34	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Ubi7, Cdc42tpb [AKT pathway], Mpig6b, Eefsec, Adgrg1, Top3b, Megf11, Mdfi, Lat, Malsu1 [Mitochondrial markers], Clns1a, Tmed4, Na8f3, Gm1979, Rrm2, Al597479, Them4, Tmem147
28.0-30.0 kDa	1290	5	40	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Krtap4-27, Crybb1, Mff [Mitochondrial markers], Kcnj11, Vps11, Pdyn, Ncor1, , Hoxb9, Cecr2 [Histone family], Pdc, Cdk4 [Cell-cycle CDK/cyclin], Slit2, Coro1b, Fubp3, Rpl6, Plk3ip1 [Actin family], Hdhd3
30.0-32.0 kDa	1101	2	24	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Svs3b, Snai2, Arhgap36, Eif4a3l1, Ppip5k1, Pdcd2l, Fbxw13, Chid1, Gm9758, Rchy1, Fitm2, Mapre1, 2610028H24Rik, Foxr1, 1700012B07Rik, Ankrd55, Mtmr11, Mmp7
32.0-34.0 kDa	1067	2	20	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Zmat3, Jmjdg6, Gjb1, C1qtnf1, Oser1, Dgkb, Nlk-1-2, Ttpa, Fcho2, D17h653e, Sh2d6, Usf1, Nup85, Btbd19, Eci1 [Mitochondrial markers], Dnae1, Mrgprg, Ccser2
34.0-36.0 kDa	2298	3	25	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Or4b1, Cd276, Slc39a3, Nat1 [Lamin family], Pabpn1, C1qtnf2, Spry1, Tomm40l [Mitochondrial markers], Dhrrs1, Or4c127, Smarca2, Fbh1, Ogn, Pex26, Or1j0, Ceacam12, Psmc3, Atad1 [Mitochondrial markers]
36.0-38.0 kDa	1168	0	17	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Vmn1r193, V1ra11, Ankrd1, Ica1, Or2aj6, Or5al5, Frzb, Or9k7, Prss32, Vmn1r84, Spdya, Kcmf1, Tas2r122, Or4a27, Vmn1r227, Septin12, Ivlb, Or14j6
38.0-40.0 kDa	918	1	30	Actin family; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	high	Spata6l, Elav2, Rcn3, Tlc3, Mapk3 [MAPK cascade], Taar8b, Nol4, Pax4, Ncoa4, Cobl11, Borcs6, Cts6, Opcml, Rnf167, Ptgrr2, Rnf39, Gmpr2, Wdr45b [Actin family]
40.0-42.0 kDa	928	1	27	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Mbd4, Tead1, Tada2b, Ptgdr, Sstr5, Lhx2, Hsd17b7, Agbl4, Sp6, Plek2, Pofut1, Lgals9, Nans, Il10rb, Cxcr1, Lgals9, Gpr27, Rpf1
42.0-44.0 kDa	869	0	34	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Zranb1, Ppp4r1, Actb12 [Actin family], Marveld3, Acta2 [Actin family], Foxi3, Art3, Opn5, Actc1 [Actin family], Septin1, Syne4, Gna11, Ttl5 [Tubulin family], Rnf34, Yy2, Hsd3b3, Bend2, Fut9
44.0-46.0 kDa	803	0	20	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Foxh1, Mtg2 [Mitochondrial markers], Mterf1b [Mitochondrial markers], Serbp1, Vsig1, Rptor, Amt [Mitochondrial markers], Anknf1, Lhx3, Fnta, Clasp2, Npy1r, Vsig1, Mapk8 [MAPK cascade], Cipc [Actin family], Wnk1, Mett14, Gm20509
46.0-48.0 kDa	840	1	36	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Sdc3, Serpina1a, Tgfb1l1, Apob, Ccdc71, Aadacl2fm2, Mppe1, Ncf1, Cers4, Etv6, Usp30, Nl5dc2, Evi, Slc17a8, Abraxas1, Pdk2 [Mitochondrial markers], Sec144a, Spata6l
48.0-50.0 kDa	742	3	21	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Mapk10 [MAPK cascade], Smoc1, Dph1, Chsi9, Prg4, Lipc, Cse1l, Tead4, Sbf1, Rnf150, Ccdc83, Kynu, Amdhd2, Kcnj16, Enr1, U2surp, Krt23, Dok3
50.0-52.0 kDa	738	3	22	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Ash10, Serpina3f, Ccdc91, Rundc3a, Adss2, Zfp113, LOC102638047, Gm38431, Cmpk2 [Mitochondrial markers], Npnt, Cdc25c, Tuba8 [Tubulin family], Ikbbk, Eef1g, Nop58, Zfp36l2, Tom112, Errf1
52.0-54.0 kDa	748	0	20	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Chia, Serpinc1, Fez1f, Immt, Cpsf7, Retreg2, Htr7, Zfp874a, Gsdma3, Krtap16-1, Rmdn3, Fam161a, Agt, Ntrk2, Tek4, Wrap73, Atg4c, Srx8
54.0-56.0 kDa	764	1	20	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Zfp810, Tmem200a, Etv4, Nab1, Ralgsps2, Tmem161a, Pnllpp2, F10, Ltv1, Vmn2r3, Polr1e, Mapk10 [MAPK cascade], Vmn2r3, Pnma3, Krt24, Slc7a11, Scarb2, Fam114a2
56.0-58.0 kDa	754	2	19	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; p53 pathway	high	Usp14, Creb5, Ifrs, Klc3, Btk, Cyp2c70, Cyp2c66, LOC620551, Praelm4, Traf2, Gcm2, Slc18a1, Slc43a3, Antxr1, Nedd4, Rnf217, Cyp2f4, Gcm2

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
58.0-60.0 kDa	645	1	16	Actin family; Cell-cycle CDK/cyclin; ER markers; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Cct6a, Ccnl2 [Cell-cycle CDK/cyclin], Fto, Clca3a1, Cyp19a1, Eefsec, Espn, SiglecF, Cyp3a59, Btrc, Slc7a1, Igf2bp2, Crnn, Aqglg1, Gm10778, Lao1, Khlh33, Ccl4
60.0-62.0 kDa	615	0	9	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	Fmo5, Camk2d, Camk2d, Ugt1a9, Lztr1, Vmn2r58, Slc22a6, Mon1b, Slc16a2, Tent4a, Trak1, Nxp65, N4bp3, Tmem184c, Nxp65, Ugt1a1, Dok7, Aggl1
62.0-64.0 kDa	545	0	9	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	C9, Mgat3, Tshz2, Slc6a20a, Vmn2r108, Lmod2, Brd4, Prkaa2, Coasy, Ntn3, Slc1a2, Ablm1 [Actin family], Pigz, Syt14, Slc22a27, Mgat3, Slc44a4, Gpr107
64.0-66.0 kDa	508	0	15	Actin family; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	Arlh1, Nr5a2, Zfp870, Rufy4, Pfkfb2, Galnt14, Hdac9 [Histone family], A330040F15Rik, Slc15a3, Espz2, Acin1, Cdkl2 [Cell-cycle CDK/cyclin], Zfp960, Fzd2, Tle6, Rbm47, Zwilch, Znf354c
66.0-68.0 kDa	482	2	11	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	high	Gm8113, Bank1, Zfp345, Zbtb22, Zbtb44, Dcaf8, Fbxw7, Frs1, Gga2, Rya, Plk5, Lrnf3, Btbd9, Hrob, Sfrk, Yrs1, Plcd4, Vmn2r105
68.0-70.0 kDa	438	1	11	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	elevated	Cry1, Rnpep, Repin1, Lrriq4, Nploc4, Ddx55, Ripk1 [Actin family], Kpna4, Cbfa2t3, Ptpn11, Prodh [Mitochondrial markers], Ccdc63, Lrrc40, Tesk1, Glg1, Gab1, Men1, Cacnb2
70.0-72.0 kDa	433	2	6	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	Mak, Chmt, Plk3, Eng, Mbd1, Zfp997, Acox1, Ncbp3, Nop9, Slc6a4, Slc23a2, Zmym5, Hspa1a [Heat shock proteins], Mak, Mtss2, Rantbp10, Glce, Sass6
72.0-74.0 kDa	398	1	8	Actin family; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; Mitochondrial markers; p53 pathway	elevated	Mpp4, Entrep3, Tacc2, Arhgap33os, Raver2, Cfap100, Gbp4, Gm12258, Mecpe, Trim32, Hnmpm, Mpp4, Anks3, Rad21, Zmynd11, Cd44, Fem1aa, Prc1
74.0-76.0 kDa	381	1	5	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	Dlg4, Fer, Fam217a, Ckap2, Foxp4, Abcg1, Duxf3, Xpr1, Zfp64, Ptptr, AU018091, Tllf12 [Tubulin family], Hdc, Zbtb33, Sass6, Pira13, Dytin, Slc26a10
76.0-78.0 kDa	372	0	10	AKT pathway; Actin family; Heat shock proteins; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	elevated	C1pb [Mitochondrial markers], Foxred2, Wdr48, Atf6b, Gramd1c, Rad50, Mcmdc2, Syt14, Zfp976, Zfp53 [p53 pathway], Sirt1, Ccdc30, Capn13, Tmcc1, Tubgcp4 [Tubulin family], Lrch1, Ppp2r1b, Pwpp3a
78.0-80.0 kDa	354	0	7	Actin family; Apoptosis caspase/BCL; Lamin family; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR	elevated	Adarb1, Brd7, Ith3, Acsf6, Cfap46, Abxn711, Zcchc8, Thop1, Thop1, Pfp1, Ikbke [NF-kB/Rel], Fbln1, Acsf1, Cdk2, Rnf19b, Med14, Mapt, Tnfrap2
80.0-82.0 kDa	354	0	6	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	elevated	Wdr49, R3hdm2, Kih4, Eps8l1, Yme1l1, Zfp111, Nexn, Gp1ba, Papola, Slc44a1, Sh3r2, C1ra, Nol10, Nup93, Elf2b5, Fancb, Trcn, Slc44a2
82.0-84.0 kDa	339	1	6	AKT pathway; Actin family; Apoptosis caspase/BCL; Heat shock proteins; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	Rpgr, Phtf1, Erc1 [Actin family], Spg7, Ints10, Map7, Mark2, Top6bl, Gfm2 [Mitochondrial markers], Agbl2, Zdhhc8, Cog2, Mthfd1, Man1a, Vps52, Kcnc3, Kif15, Kri1
84.0-86.0 kDa	302	2	7	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	Nasp, Zfp184, Dhx16, Lima1 [Actin family], Tle1, Dtna, Dtna, Fam83c, Sox5, Rps6ka6, Sytl5, Sox5, Arhgap24, Utrn, Cdh24, Phtf2, Zranb1, Arhgap24
86.0-88.0 kDa	314	0	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	Rps6ka6, Zgrf1, Hgs, Fhip2a [Actin family], Aoc1, Itgb6, Zfp458, Map7d2, Enah, Snap91, Sgip1 [Actin family], Cog7, Fam193b, Cog7, Misd6, Sh3d19, Supti20, Axin2
88.0-90.0 kDa	245	0	5	Actin family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; p53 pathway	elevated	Farp1, Stat1 [JAK-STAT], Kcnq2, Vac14, Itgb5, Stat3 [JAK-STAT], Grip2 [Actin family], Unk, Zw10, Brd2, Trpv1, Acap2, Pkp2, Cdh11, Dact1, Pkp4, Top3a, Zly2
90.0-92.0 kDa	252	0	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Nuclear receptor PPAR; Tubulin family; p53 pathway	elevated	Stat5b [JAK-STAT], Gapvd1, Arhgap12, Prom1, Zfx, Elfn2, Cptlc1, Adam8, Arhgef5, Kcnq5, Fyb1, Rasa4, Itgb5, Synm, Adamts4, Septin3, Taf4b, Matn2
92.0-94.0 kDa	262	0	7	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	Poln, Exo1, Taf1c, Avil, Cdc5lrf6, Fam171b, Trim71, Arhgap26, Abxn21, Cdc5lrf7, Adam9, Z310069G16Rik, Cdh3c, Col9a1, Col9a1, Klf9, Txndc16, Tsr1
94.0-96.0 kDa	250	1	3	Actin family; Apoptosis caspase/BCL; Heat shock proteins; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	Tbc1d5, Cdhrl1, Dnm1, Wdr75, Exoc6, Zbtb10, Vmn2r94, Mical1, Zfp110 [Actin family], Grik1, Cfh, Lrrc8a, Cacna1f, Exoc6b, Prickle1, Hspa4 [Heat shock proteins], Esys2, Vmn2r13
96.0-98.0 kDa	331	0	14	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	Hnmpu11, Polk, Gm61617, Armc3, Ilf3, Pdc6ip [Actin family], Znf629, Grm4, Tct7b, Sycp2, Ankrd28, Map3k12 [MAPK cascade], Skor1, Kctd1, Kctd1, Xrra1, Vmn2r15, Kansl3, Vmn2r22
98.0-100.0 kDa	269	0	3	Actin family; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers	elevated	Dpp9, Ctndd1, Whrm, Trpv4, Esf1, Vmn2r117, Vmn2r109, Dnm2, Unc5d, Rtk1, Lrrc8d, Exosc10, Aco1, Hps3, Fycol, Vav1, Atp6v0a2, Dnm2
100.0-102.0 kDa	233	0	5	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	elevated	Nfatc2, Cdh4, Sncap, Zmym1, Eysy3, Clca1, Pld1, Prkd3, Atxn2, Chsy3, Spata31e5, Hk3, Cttna1, Clca3a1, Ddias, Dlg1, Rapgef3, Npepps
102.0-104.0 kDa	235	0	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	Lpin1, Rnf31, Ganc, Prkd1, Lpin1, Cip2a, Slc14a2, Pcdhga6, Pcdha10, Clca4a, Mcm2, Ctndd1, Tsc22d1, Snd1, Eif4g2, Eif4g2, Hace1, Adgre1
104.0-106.0 kDa	203	0	6	Actin family; Cell-cycle CDK/cyclin; JAK-STAT; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	Mrtfa, Ap2a2, Dgkz, Grik3, Ror1, Arhgap4, Ctndd1, Thbs3, Lig4, Unc5d, Dgkz, Lgr4, Msh2, Ripor3, Taf4, Tnpo3, Sik2, Sftswap
106.0-108.0 kDa	169	0	5	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers	elevated	Bclaf1 [Apoptosis caspase/BCL], Ptpn, Gm61616, Atp13a5, Mdga1, Pcsk6, Phc1, Nbr1, Kdm7a, Reck, Rbm19, Ptpn, Alosa, Cnrm1, Mecom [Histone family], Grin1, Grik1, Lig3
108.0-110.0 kDa	183	0	0	Actin family; Cell-cycle CDK/cyclin; Histone family; p53 pathway	elevated	Iqsec1, Pard3, Grin1, Fhip1b [Actin family], Slc8a1, Dlgap4, Slc8a1, Drp2, Myo19, Ttc17, Chtf18, Chtf18, Als2cl, Ctndd1, Thrapp3, Ints4, Fbrsl1, Chtf18
110.0-112.0 kDa	165	0	3	Actin family; Histone family; MAPK cascade; Mitochondrial markers	elevated	Cenpf, Emf4, Srx14, Usp37, Epha3, Usp37, Septin4, Osmr, Ccdc39, Mib1, Vps8, Ephb6, Nbr1, Map3k21 [MAPK cascade], Atp11b, Plk3ca, Rtel1, Merlk
112.0-114.0 kDa	162	1	7	Actin family; Apoptosis caspase/BCL; Histone family; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	elevated	Syne3, Sipal1, Ppargc1b [Nuclear receptor PPAR], Plekhm2, Srx25, Rad51ap2, Rasa13, Grid1, Atp1a2, Nedd4f, Ubn2, Sipal1, Krtf78, Grip2 [Actin family], Gltf2, Rbm44, Corin, Usp15
114.0-116.0 kDa	145	0	3	Actin family; Apoptosis caspase/BCL; Histone family	elevated	Popl1, Jarid2, Atp4a, Crim1, Pdzm4, Smg5, Pum2, Mrip1 [Actin family], Ano2, Rbm26, Col19a1, Cd101, Sreb1f, Tshz2, Fkbp15, Dgkh, Kdm4b [Histone family], Crybg3
116.0-118.0 kDa	151	1	1	Actin family; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	Zfp616, Nlrp9a, Tdrd5, Nebl, Nup210 [p53 pathway], Ogdh, Ankrd28, Grip1 [Actin family], Taok1, Ipo9, Megf11, Plcb4, Myo1d, Tnk2, Usp38, Pum1, Ephae6, Ccdc57
118.0-120.0 kDa	149	0	7	AKT pathway; Actin family; Apoptosis caspase/BCL; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	elevated	Sk11ip [Actin family], Frmd4b, Ryr2, Kif11, Uba1y, Pelp1, Zfp827, Specc1, Nlrc4, Sugp2, Zbtb21, Limch1, Ube4a, Rimbp2, Igch, Chd8, Tmrss9, Kcnh3
120.0-122.0 kDa	123	0	4	Actin family; Histone family; MAPK cascade	elevated	Ank2, Plk3cd, Tek, Znf451, Asap1, Rab11fip1 [Actin family], Arhgap30, Atp13a2, Cep192, Plk3cd, Zfp729b, Gemin4, Kib, Ccdc136, Fhip1a [Actin family], Cep131, Mrc2, Vps13d
122.0-124.0 kDa	137	1	4	Actin family; Histone family; JAK-STAT; Lamin family	elevated	Grip1 [Actin family], Fmnl1, Pde3b, Nos3, Tmem132c, Anknf1, Trhd, Pde3b, Itga6, Tdrd7, Rapgef1, Adamts2, Bnc2, Abca4, Ubn1, Dsg2, Col16a1, Guicy2d
124.0-126.0 kDa	117	0	2	Actin family; Histone family; Tubulin family; p53 pathway	elevated	Phkb, Arhgap45, Rantbp17, Mamdc4, Srgap1, Wwc1, Tacc2, Cp, Mybpc3, Spata31e1, Amer1, Adamts17, Slc4a5, Sh3pdx2a, Cp, Wdhd1, Adnp, Wdhd1
126.0-128.0 kDa	128	0	5	Actin family; p53 pathway	elevated	Trp53bp2 [p53 pathway], Pcdh19, Tdrd12, Nek10, Adgrl2, Rfc1, Kcnma1, Catsperb, Kank1, Agbl1, Pcdh17, Agbl1, Bmp2k, Mybpc1, Pbrn1, Btbd7, Lats1, Slc12a5
128.0-130.0 kDa	134	0	3	Actin family; Histone family; Lamin family; Mitochondrial markers	elevated	Obsl1, Iqsec2, Mybpc1, Sorcs1, Tnks2, Rapgef6, Abl2, Ccdc88b, Polr1b, Abl2, Trappc9, Syngap1, Grm5, Cfap70, Itgal, Dhx37, Erc1 [Actin family], Erc1 [Actin family]
130.0-132.0 kDa	114	0	5	Actin family; Apoptosis caspase/BCL; Histone family; JAK-STAT; Lamin family; NF-kB/Rel	elevated	Uty, Astn1 [Actin family], Ralgapb, Kif1b, Tbc1d8, Hipk3 [Actin family], Dgkh, Vwa3a, Rbm20, Fnipl [Actin family], Fbf1, Irs4, Lamc2 [Lamin family], Zmynd8, Nos2, Nfasc, Sgo2b, Hr
132.0-134.0 kDa	121	0	4	Actin family; Apoptosis caspase/BCL; Histone family; JAK-STAT; Tubulin family	elevated	Camta2, Rapgef1, Grid2ip, Myo1b, Coar1 [Apoptosis caspase/BCL], Atp2b2, Adgrl2, Kidins220 [Actin family], Nfasc, Dnaaf9, Pcdh9, Phka1, Prx, Clip, Sorbs2, Glg1, Catsperg1, Ripor1 [Actin family]
134.0-136.0 kDa	133	0	8	Actin family; Apoptosis caspase/BCL; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	Ttc3, Ctcl, Card11 [Apoptosis caspase/BCL], Wapl, Mn1, Kcnma1, Washc5, Dgkh, Per1, Zbtb38, Nlrp1b, Jag1, Lars1, Zmym4, Hdlbp, Smc2, Sorts2, Nlrp1a
136.0-138.0 kDa	110	0	1	Actin family; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	Zc3n4, Nbeal1, Slc4a2, Mmm1, Plekhg1, Aft3, Alpk1, Pcdh9, Lcor, Sorbs2, Gse1, Adamts16, Plch2, Slc4a7, Cand1, Nphs1, Spem3, Aut52
138.0-140.0 kDa	92	0	0	Actin family; Histone family; MAPK cascade; NF-kB/Rel	routine	Dzip3, Dzip3, Col1a1, Ehm2 [Histone family], Brpf1, Spata13, Adgr3, Mecom [Histone family], Impg2, Kcnt1, Zbtb40, Nckap5l, Col15a1, Csppl1, Kcnma1, Inf2, Plcb1, Sh3tc2
140.0-142.0 kDa	106	0	0	Actin family; Histone family; MAPK cascade; Tubulin family; p53 pathway	routine	Adcy8, Stag1, Ppfia1, Af2, Abcc4, Vars1, Sall1, Sall1, Crybg3, Slc4a7, Tbc1d4, Ktn1, Ppip5k2, Abcb4, Diaph1, Llicam, Ankar, Col15a1
142.0-144.0 kDa	74	0	4	Actin family; Cell-cycle CDK/cyclin; MAPK cascade; Tubulin family	elevated	Adgrl3, Rasa2, Tbc1d1, Psd3, Arid4a, Gm382, Zmym3, Zfp638, Ulk4, Gm20547, Arhgap29, Clasp2, Mast3, Dhx8, Sympk, Cdh2, Adgrl3, Ppfia1
144.0-146.0 kDa	98	0	3	Actin family; Histone family; MAPK cascade; Tubulin family	elevated	Skf1, Dop1b, Rasgrf1, Mypn, Dennd5a, Mast3, Sik36, Smc4, Mapk8ip3 [Actin family; MAPK cascade], Ccdc40, Tnik, Ctnnap5b [Actin family], Crb1, Nek1, Iars1, Plekha5, Pxdn, Pcthl
146.0-148.0 kDa	87	0	3	Actin family; MAPK cascade; Tubulin family	elevated	Cluh, Plekhg2, Adgrl3, Spata31f1b, Spata31f1c, Spata31f1d, Spag9 [Actin family], Usp19, Myo6, Srcin1, Usp31, Zlyve9, Ktn1, Synj1, Rims2, Usp19
148.0-150.0 kDa	76	0	2	Actin family; Apoptosis caspase/BCL; Lamin family; MAPK cascade	routine	Trim66, Clip1, Ptpnm2, Tbc1d32, Cluh, Myo3b, Myo6, Cgnl1, Ctnnap2 [Actin family], Syngap1, Ctnnap2 [Actin family], Kank1, Aox4, Znf541, Zfp827, Myo6, Zfp644, Disp3

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
150.0-152.0 kDa	76	0	0	Actin family	routine	Klf16b, Phldb1, Mink1, Ptpfr, Nup155, Myof, Klf24, Pds5a, Nphp3, Trnc6a, Pds5a, Ncor1, Nfasc, Trnk [Actin family], Mink1, Carmi3, Kdr, Pnpla7
152.0-154.0 kDa	56	0	2	Actin family; Histone family; MAPK cascade	routine	Nup153, Tns2, Mybpbp1a, Ube4b, Palld, Npat, Cracd [Actin family], Nsd2 [Histone family], Ccdc171, Cpd, Zlyve9, Atrnl1 [Actin family], Edc4, Edc4, Lrrc9, Rab3gap2, Rab3gap2, Ktn1
154.0-156.0 kDa	52	0	1	Actin family; MAPK cascade	routine	Trnk [Actin family], Pogz, Spata31e2, Scaf11, Nirc5, Kdm6a, CEFIP [Actin family], Espn, Map3k5 [MAPK cascade], Rab3gap2, Zmym2, Iqcn, Cux2, Kmt2a, D5Erd579e, Synj2, Kiaa0232, Tin2
156.0-158.0 kDa	38	0	1	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers	routine	Brd4, Arhgef28, Med23, Nsd3 [Histone family], Ctnnap1 [Actin family], Pogz, Cux1, Med23, Mia2, Rgs12, Rprd2, Mia2, Lrrprc [Mitochondrial markers], Lrrc7, Bcl9 [Apoptosis caspase/BCL], Nacad, Z700049A03Rik, Btlp3a
158.0-160.0 kDa	36	0	3	Actin family; Cell-cycle CDK/cyclin	routine	Nfat5, Atrn [Actin family], Rock1, Ssh2, Nup160, Atg10d, Doc, Blm, Rrbp1, Ncoa2, Synj2, Plin4, Phldb1, Prrc2b, Mlh3, Naip1, Blm, Ssh2
160.0-162.0 kDa	70	0	4	Actin family; Histone family; MAPK cascade	elevated	Ctip1, Ylpm1, Nos1, Kif15, Clasp1, Mbd5, Prrc2c, Nos1, Smg6, Lmtk2, Adgrb2, Rock2, Tex16, Rock2, Col4a1, Zfp462, Ctnln, Ilt140
162.0-164.0 kDa	58	0	4	Cell-cycle CDK/cyclin; MAPK cascade	elevated	Alp7a, Scaf11, Fancd2, Clasp1, Arhgap23, Arap1, Adgrl1, Fycoo1, A2ml1, Znf608, Prx, Mug2, Gapvd1, Ptpnu, Asx1, Mlh3, Tex14, Cep164
164.0-166.0 kDa	69	0	1	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers	routine	Col4a6, Naip2, Col4a4, Plekhg1, Crybg2, Ptpkr, Nrxn1, Cux1, Plch2, N4bp2, Nos1, Camsap2, A2m, Clasp1, Ptprr21, Adgrl2, Pds5b, Pcf11
166.0-168.0 kDa	61	0	1	Actin family; Lamin family	routine	Ptpkr, Camsap2, Nrxn1, Arhgef40, Ralgapb, Prex1, Patj, Mtlc1, Rapgef2, Rapgef2, Atp10b, Adgrl1, Gemin5, Ralgapb, Adgrl2, Gemin5, Ccdc88b, Kcp
168.0-170.0 kDa	45	0	2	Actin family; Histone family; MAPK cascade	routine	Clasp1, Shoc1, Ahdc1, 4930407110Rik, Tns3, Ptprr, Adgrb2, Robo2, Pbrm1, Limch1, Adgrl3, Slti2, Atp10a, Rb1cc1, Resf1, Topbp1, Dcaf1, Kmt2c [Histone family]
170.0-172.0 kDa	37	0	1	AKT pathway; Actin family	routine	Eprs1, Disp1, Dst, Sli2, Aqr, Arhgap35, Pla2r1, Dip2a [Actin family], Baz1b [AKT pathway], Madd, Adgrl3, Cep170b, Dip2c [Actin family], Trpm1, Clasp1, Mast1, Dip2c [Actin family], Adgrl3
172.0-174.0 kDa	46	0	0	Lamin family	routine	Ankrd31, Sycp2, Cdc42bpg, Trpm2, Lamc3 [Lamin family], Zfp646, Arhgef12, Slx4, Zmym4, Arhgap5, Arhgef12, Cacna1s, Adgrl3, Smarca2, Akap12, Synj1, Madd, Nhsl1
174.0-176.0 kDa	35	0	0	Nuclear receptor PPAR	routine	Smarca2, Zfp831, Abcc9, Agl, Ralgap2a, Madd, Adgrl3, Duox2, Dlg5, Madd, Eif4g3, Eif4g3, Eif4g3, Alk, Ptprr21, Rims2, Nisch, Cep170
176.0-178.0 kDa	34	0	0	Actin family; Lamin family	routine	Camsap1, Eif4g1, Hecw2, Synj1, Kdm6b, Robo1, Robo1, Ugg1, Madd, Plxnc1, Ino80, Fhod3, Katnip [Actin family], Nhs [Actin family], Usp54, Arhgef5, Kdm5d, Dnm1b
178.0-180.0 kDa	26	0	1	Actin family; MAPK cascade	routine	Abcc8, Baz1a, Myo15b, Pclo, Baz1a, Vcan, Ylpm1, Ltbp4, Dennd4c, Ctnnbp2 [Actin family], Lrp5, Cfp474, Peg3, Nhs [Actin family], Rapgef6, Klf26b, Thsd7b, Hecw1
180.0-182.0 kDa	39	0	0	Actin family	routine	Erich3, Alpk3, Samd9l, Trnc6c, Trpm3, Rapgef6, Smarca2 [Actin family], Cnot1, Lrp6, Lrp6, Tet3, Depdc5, Mroh2b, Iqgap2, Ankrd26, Bcor1, Akap12, Smarca2
182.0-184.0 kDa	36	0	1		routine	Sphkap, Pbrm1, Col18a1, Samd9l, Col18a1, Adamts7, Rb1cc1, Phlpp1, Unc13b, Ttc3, Als2, Thoc2l, Smarca2, Thoc2, Rims1, Siglec1, Kif21b, Gcc2
184.0-186.0 kDa	41	0	1	Actin family	routine	Nrxn2, Phr1, Rapgef2, Rreb1, Pik3c2b, Abca8a, Adprh1l, Wiz, Camta1, Trpm1, Wiz, Tut4, Mdc1, N4bp2, Myo3a, Crybg1, Mdc1, Ehbp1l1
186.0-188.0 kDa	28	0	0	Histone family	routine	Setd1a [Histone family], Ncor1, Kif21b, Col27a1, Plec, Adcy10, Kif14, C3, Eif2ak4, Eif2ak4, Kif21a, Myo18a, Crocc2, Ltbp1, Ptprr, Cacna1d, Ltbp1, Dst
188.0-190.0 kDa	26	0	0		routine	Argef3, Mon2, Adgb, Mast2, Atg2a, Iqgap1, Itsn2, Tjp1, Btlp1, C5, Afdn, Itsn2, Tasor2, Mroh2a, Sipa1l1, Mon2, Fndc1, Syne2
190.0-192.0 kDa	34	0	2		routine	Mroh2a, Dst, Bcor1, Arhgef28, Abca15, Mast2, Abca15, Ltbp2, Gcc2, Pik3c2a, Pik3c2a, Btbd8, Kif1a, Peak1, Dop1a, Casz1, Arid1b, Ank3
192.0-194.0 kDa	36	0	0	Histone family	routine	Adamts1, Kif1a, Marf1, Scn7a, Kdm5a, Ankrd26, Abca16, Abca14, Cul7, Lmo7, Tiam2, Bcor, Cdc42bpa, Z900026A02Rik, Col14a1, Adamts1, Ltbp2, Eif4g3
194.0-196.0 kDa	40	0	1	Actin family	routine	Tsc2, Tsc2, Polr1a, Lmo7, Fndc1, Nckap5, Itsn1, Xrn1, Xrn1, Qser1, Dmbt1, FAM186A, Gcc2, Trip11, Tns1, Pbrm1, Tjp1, Cdc42bpb
196.0-198.0 kDa	36	0	0	Actin family; Lamin family; Tubulin family	routine	Abca17, Tubgcp6 [Tubulin family], Klf26a, Strc, Chd1, Cep152, Lamb2 [Lamin family], Xirp1 [Actin family], Nup188, Zfp646, Cdc42bpa, Sipa1l1, Zswim8, Lamb1 [Lamin family], Tubgcp6 [Tubulin family], Tns1, Nup98, Nup98
198.0-200.0 kDa	23	0	1	Actin family; p53 pathway	routine	Mast2, Trnc6c, Patj, Cdc42bpa, Brca1, Pappa2, Tsc2, Ltn1, Focad, Tns1, Rtl1, Suptf6h, Map2, Ubr2, Kidins220 [Actin family], Nup210 [p53 pathway], Rbbp6, Rp1l1
200.0-202.0 kDa	18	0	0	Lamin family	routine	Tanc1, Shank2, Ubr1, Nalcn, Tns1, Togaram1, Nalcn, Myo15a, Tanc1, Patj, Nckap5, Tsc2, Tns1, Scn11a, Virma, Rai1, Itgb4, Lama4 [Lamin family]
202.0-204.0 kDa	21	0	0	Lamin family	routine	Tsc2, Kif1a, Argef2, Nav1, Lamb1 [Lamin family], Ank1, Myo5c, Zfp236, Brwd3, Ank1, Wdr90, Nyrnln, Trnc6a, Kif20b, Atad5, Zc3h13, Ppl, Parp14
204.0-206.0 kDa	22	0	1	Cell-cycle CDK/cyclin; Histone family; p53 pathway	routine	Ppl, Kif1b, Nup210 [p53 pathway], Ank1, Ank1, Ank1, Aldn, Kmt2e [Histone family], Kif13b, Tmem131, Eif4g3, Cntrl, Kif13b, Agrn, Wdr90, Sbf2, Cacna1i, Shroom3
206.0-208.0 kDa	24	0	0	Actin family; p53 pathway	routine	Tro, Ankrd31, Znf142, Plxn2b, Rlf, Arhgef4, Aldn, Notch4, Phip [Actin family], Phip [Actin family], Gbf1, Ank1, Ank1, Btaf1, Nes, Virma, Pcm1, Lcorl, Aldn
208.0-210.0 kDa	28	0	0	Histone family; p53 pathway	routine	Rosl, Tchh, Tlcr, Plxn3b, Argef1, Kat6b [Histone family], Mgam, Sbf1, Nup210l [p53 pathway], Nup210l [p53 pathway], Akap11, Scn4a, Scn4a, Zfp106, Znf106, Sis, Nckap5, Cfpaf5
210.0-212.0 kDa	34	0	0	Actin family	routine	Cacna1s, Ralgap2a, Cacna1s, Hcfc1, Sbf2, Taf1, Myo5b, Myo5b, Chd2, Atg2a, Plxna1, Wnk2, Psm4, Myo16, Dennd4c, Ptprr, Plxna2, Cfpaf4
212.0-214.0 kDa	26	0	1	p53 pathway	routine	Adamts20, Dlg5, Tet2, Trpm7, Myo5a, Trpm7, Plxna4, Tp53bp1 [p53 pathway], Ubr3, Mylk, Nup214 [p53 pathway], Tet2, Skt, Dmbt1, Tic6, Myo5b, Vwa8, Polr2a
214.0-216.0 kDa	26	0	0	Histone family; p53 pathway	routine	Ank3, Ralgap2a, Muc12, Shroom3, Dlg5, Ptprr, Taf1, Dock5, Zfp638, Taf1, Pcdh15, Emf5, Shroom3, Hcfc1, Dock1, Setd1b [Histone family], Gm45861, Myo5a
216.0-218.0 kDa	20	0	0	p53 pathway	routine	Tdrd6, Parp4, Map2, Chd4, Arhgap21 [p53 pathway], Agrn, Dicer1, Dicer1, Ank3, Greb1, Arhgap21 [p53 pathway], Polr2a, Thada, Mcm3ap, Rlf, Ank3, Exph5, Chd4
218.0-220.0 kDa	22	0	1	Actin family; Apoptosis caspase/BCL; Histone family	routine	Mical2 [Actin family], Znf638, Zfp638, Dock11, Eif4, Chd4, Wnk2, Ckap5, Mpdz, Znf1x1, Brd10, Chd5, Ralgap2a, Casp8ap2 [Apoptosis caspase/BCL], Tet1, Prrc2a, Ak9, Ncoa6
220.0-222.0 kDa	20	0	0	Actin family	routine	Emf5, Mpdz, Tanc2, Naca, Scn10a, Scn10a, Trip12, Scn8a, Mical3 [Actin family], Scn3a, Cacna1f, Myh7b, Pcnx3, Arhgef17, Cacna1c, Cacna1f, Myh15, Wnk2
222.0-224.0 kDa	27	0	0	Actin family	routine	Unc13b, Wnk2, Dscam, Chd5, Dnah9, Dock4, Chd5, Myh8, Myh4, Myh7, Tet1, Cacna1c, Myh2, Myh1, Myh11, Triobp [Actin family], Crybg1, Ckap5
224.0-226.0 kDa	32	0	0	Actin family; Histone family	routine	Trip12, Fer14, Dscaml1, Cacna1h, Heatr5b, Ptprr, Scn9a, Kat6a [Histone family], Kat6a [Histone family], Zfc3h1, Scn9a, Scn8a, Scn8a, Scn8a, Ubr4, Shank1, Shank1, Lrrk1
226.0-228.0 kDa	27	0	0		routine	Fancm, Otof, Wnk2, Scn8a, Shank1, Myh9, Trip11, Ccdc88c, Dock4, Gtf3c1, Svll, Dmbt1, Crocc, Otof, Myh11, Otof, Myh11, Wnk2
228.0-230.0 kDa	10	0	0	Actin family; Histone family	routine	Kat6b [Histone family], Myh14, Scn1a, Pcm1, Chd3, Myh10, Prrc2a, Ralgap1, Mical3 [Actin family], Arhgap32
230.0-232.0 kDa	14	0	0	Actin family; Histone family	routine	Cit [Actin family], Dscaml1, Svll, Macf1 [Actin family], Kat6a [Histone family], Myo18a, P4ka, Plxn1, Atg2b, Gm19410, Ccdc88a, Cacna1c, Tnc, Cit [Actin family]
232.0-234.0 kDa	19	0	0		routine	Evpl, Pikfyve, Myo18a, Myh10, Chd3, Myo18a, Trpm6, Dock3, Ankrd12, Esp1l, Nup205, Sytl2, Dock6, Dock3, Alpk2, Myof, Myof, Myh10
234.0-236.0 kDa	21	0	0	Actin family	routine	Baz2b, C2cd3, Wnk2, Pcnx2, Ralgap1, Lrba, Rp1, Ralgap1, Myof, Dock9, Frem3, Ccdc187, Dock9, Dock9, Cit [Actin family], Cit [Actin family], Frem3, Numal
236.0-238.0 kDa	23	0	0	Actin family	routine	Malrd1, Cad, Pkd13, Dock6, Pikfyve, Abca7, Cad, Arid1b, P4ka, Cit [Actin family], Loxhd1, Dysf, Spel2, Myo10, Dysf, Gtf3c1, Usf3, Abca7
238.0-240.0 kDa	19	0	0		routine	Tdrd6, Tdrd6, Dock7, Knl1, Dock7, Dock9, Med13, Myo9b, Dock8, Dysf, Dysf, Sytl2, Dysf, Dysf, Ccdc187, Tecta, Cacna1c, Fn1
240.0-242.0 kDa	23	0	0		routine	Argef3, Cacna1c, Cacna1c, Cacna1c, Cacna1c, Sdk1, Myo9b, Dysf, Myo7b, Med12, Ylpm1, Dock7, Dysf, Pkdre, Dock9, Med12l, Dock7, Dock7
242.0-244.0 kDa	26	0	0		routine	Heatr1, Cacna1c, Arid1a, Ice1, Arid1b, Cacna1c, Arid1a, Med13l, Arid1b, Cacna1i, Armcv4, Cacna1c, Apc2, Prrc2b, Apc2, Svll, Cabin1, Cad
244.0-246.0 kDa	14	0	0		routine	Adgrv1, Dop1b, Notch3, Sp1bn1, Med12l, Frem1, Snrnp200, Med12, Cacna1d, Ints1, Ints1, Sp1b, Cacna1c, Zfp407
246.0-248.0 kDa	19	0	0		routine	Apc2, Cacna1c, Sp1b, Dock10, Col6a6, Znf318, Col6a6, Ankrd17, Gon4l, Potefam1, Cacna1d, Sorl1, Nav3, Dido1, F5, Hmnr, Abcf1, Syne2
248.0-250.0 kDa	14	0	0		routine	Cacna1d, Rttm, Gpr179, Gon4l, Cacna1g, Cacna1c, Cacna1g, Dmxl2, Myo7a, Fn1, Dock10, Myo7a, Cacna1g, Unc13c
250.0-252.0 kDa	11	0	0		routine	Fn1, Myo7a, Kntc1, Ascc3, Pde4dip, Cacna1g, Col6a4, Wnk1, Flg2, Cacna1g, Cacna1g
252.0-254.0 kDa	15	0	0		routine	Qrich2, Son, Cacna1g, Nav3, Cspg4, Fn1, Loxhd1, Nav3, Cacna1g, Svll, Hwep3, Tln2, Ncor2, Abca1, Cacna1g
254.0-256.0 kDa	15	0	0	Heat shock proteins	routine	Son, Sec16a, Akap6, Ptprr21, Dnajc13 [Heat shock proteins], Btlp2, Urb1, Urb1, Myo7a, Cacna1e, P1ce1, Dnajc13 [Heat shock proteins], Ptprr, C2cd3, Nav2
256.0-258.0 kDa	12	0	0	Actin family	routine	P1ce1, Potefam1, Ptprr, Potefam1, Cacna1b, Cacna1e, Cacna1e, Mprp [Actin family], Dop1b, Pcnx1, Golga4, Catspert
258.0-260.0 kDa	6	0	0		routine	Cic, Pcnx1, F8, Cacna1h, Brwd1, Ros1

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
260.0-262.0 kDa	13	0	0	Histone family	routine	Fn1, Abca4, Otolg, Cnot1, Crebbp [Histone family], Cacna1b, Cacna1h, Dsp, Cacna1b, Cacna1b, Nav2, Cacna1h, Ros1
262.0-264.0 kDa	12	0	0	Histone family	routine	Cacna1h, Pole, Cacna1a, Vcan, Cacna1g, Fn1, Spag17, Ep300 [Histone family], Cacna1a, Ankd1, Ankr17, Cacna1h
264.0-266.0 kDa	12	0	0	Histone family	routine	Ncor1, Tasor2, Mgam2-ps, Mast4, Mast4, Acaca, Crebbp [Histone family], Crebbp [Histone family], Notch2, Son, Cspg4b, Nbas
266.0-268.0 kDa	12	0	0		routine	Mast4, Cnot1, F8, Rlf1, Vps13a, Mast4, Hivp2, Cnot1, Tpr, Ttc28, Dnah12, Cacna1a
268.0-270.0 kDa	11	0	0		routine	Sptb, Cntrl, Son, Cntrl, Cep295, Dop1a, Ncor2, Ankd1, Ncor2, Tln1, Ncor2
270.0-272.0 kDa	16	0	0		routine	Dmd, Ptpn13, Map1b, Ptpn13, Abca2, Abca2, Ncor1, Kalrn, Ttc28, Ncor1, Bdp1, Notch1, Sptbn2, Tln2, Tln2, Pkd112
272.0-274.0 kDa	14	0	0	Actin family; Histone family	routine	Pkd112, Fasn, Pde4dip [Actin family], Fasn, Setd2 [Histone family], Fn1, Cep295, Ncor2, Zfhx2, Prpf8, Zdbf2, Igf2r, Spg11, Tpr
274.0-276.0 kDa	10	0	0	Lamin family	routine	Ankrd17, Ankrd17, Sptbn1, Zdbf2, Flna [Lamin family], Fcgbp, Fcgbp, Unc79, Unc79, Acacb
276.0-278.0 kDa	9	0	0	Actin family; Lamin family	routine	Stab1, Cep192, Stab1, Dop1a, Cep250, Pde4dip [Actin family], Stab2, Cacna1a, Finb [Lamin family]
278.0-280.0 kDa	5	0	0		routine	Cep295, Atrx, Cep250, Cep250, Spta1
280.0-282.0 kDa	9	0	0	Lamin family	routine	Fcgbp1, Flna [Lamin family], Polq, Gvin1, Gvin1, Flna [Lamin family], Cul9, Jmjd1c, Jmjd1c
282.0-284.0 kDa	5	0	0		routine	Cul9, Bahcc1, Znf462, Sptan1, Zfyve26
284.0-286.0 kDa	13	0	0	Histone family	routine	Mast4, Nsd1 [Histone family], Nsd1 [Histone family], Ralgapa1, Sptan1, Ank3, Lrrk2, Sptan1, Sptan1, Igsf10, Setd2 [Histone family], Trnc18, Muc16
286.0-288.0 kDa	4	0	0		routine	Lrba, Cenpe, Cenpe, Sptan1
288.0-290.0 kDa	13	0	1		routine	Cdcp3, Hivp1, Hivp1, Col6a3, Col6a3, Mtor, Sptbn4, Myo18b, Cep290, Hectd1, Col6a5, Col6a5, Cep290
290.0-292.0 kDa	11	0	0	Lamin family	routine	Hectd1, Usp9x, Usp9x, Unc79, Epg5, Pkd11l1, Chd8, Flnc [Lamin family], Tep1, Trrap, Piezo1
292.0-294.0 kDa	6	0	0		routine	Piezo1, Piezo1, Myo9a, Usp9y, Abca12, Gcn1
294.0-296.0 kDa	7	0	0	Histone family	routine	Usp24, Gm9195, Kmt2b [Histone family], Srm2, Col7a1, Adgrg4, Kmt2b [Histone family]
296.0-298.0 kDa	7	0	0	Histone family	routine	Usp24, Ankrd11, Nsd1 [Histone family], Pdxd2, Unc79, Megf8, Setx
298.0-300.0 kDa	4	0	0		routine	Prrc2c, Ankrd11, Muc5ac, Nbeal2
300.0-302.0 kDa	10	0	0		routine	Map1a, Atr, Pkd11l1, Atr, Muc6, Tenm3, Atr, Zfp292, Myo9a, Nbeal2
302.0-304.0 kDa	7	0	0		routine	Prrc2c, Cfap46, Nbeal2, Tenm3, Tenm3, Igfn1, Akap13
304.0-306.0 kDa	7	0	0		routine	Itpr3, Tg, Tenm1, Tacc2, Chd6, Csmc3, Tenm1
306.0-308.0 kDa	8	0	0		routine	Akap13, Nbeal1, Tenm2, Tenm2, Tenm2, Apc, Itpr2, Trnc18
308.0-310.0 kDa	7	0	0		routine	Ubr5, Ubr5, Tenm4, Ubr5, Vwf, Vwf, Lrba
310.0-312.0 kDa	6	0	0		routine	Prrc2c, Apc, Prrc2c, Apc, Piezo2, Tex15
312.0-314.0 kDa	5	0	0		routine	Mga, Fbn1, Itpr1, Otag, Fbn2
314.0-316.0 kDa	2	0	0		routine	Ank3, Nipbl
316.0-318.0 kDa	5	0	0		routine	Lrba, Celsr2, Lrba, Utp20, Utp20
318.0-320.0 kDa	3	0	0		routine	Piezo2, Bltp1, Nf1
320.0-322.0 kDa	2	0	0		routine	Piezo2, Bptf
322.0-324.0 kDa	2	0	0		routine	Hmr, Chd9
324.0-326.0 kDa	4	0	0		routine	Crybg3, Piezo2, Piezo2, Map1a
326.0-328.0 kDa	4	0	0		routine	Helz2, Nbea, Bptf, Bod1l
328.0-330.0 kDa	6	0	0		routine	Zzef1, Zzef1, Speg, Mga, Pcnt, Pcnt
330.0-332.0 kDa	8	0	0	Histone family	routine	Celsr1, Zzef1, Ash1l [Histone family], Pcnt, Fam186a, Dst, Gm42715, Helz2
332.0-334.0 kDa	6	0	1		routine	Dsp, Myo15b, Mga, Bptf, Srcap, Col12a1
334.0-336.0 kDa	3	0	0		routine	Helz2, Chd7, Adgrg4
336.0-338.0 kDa	6	0	0		routine	Dmd1, Kalrn, Ep400, Wdfy4, Dmx1, Fryl
338.0-340.0 kDa	6	0	0	Lamin family	routine	Lama1 [Lamin family], Dmx1, Fry, Fry, Prune2, Trnb
340.0-342.0 kDa	4	0	0		routine	Col12a1, Hmr, Dmx1, Ranbp2
342.0-344.0 kDa	4	0	1	Lamin family	routine	Cenpf, Srcap, Trank1, Lama2 [Lamin family]
344.0-346.0 kDa	2	0	0		routine	Htt, Htt
346.0-348.0 kDa	3	0	0		routine	Dchs1, Cep350, Trio
348.0-350.0 kDa	3	0	1		routine	Atm, Srcap, Atm
350.0-352.0 kDa	3	0	0		routine	Frem2, Rev3l, Mki67
352.0-354.0 kDa	3	0	0		routine	Neb, Cfap54, Col6a3
354.0-356.0 kDa	3	0	0		routine	Speg, Wdfy4, Speg
356.0-358.0 kDa	2	0	0		routine	Vps13d, Dnah12
358.0-360.0 kDa	4	0	0		routine	Celsr3, Cplane1, Celsr3, Vps13a
360.0-362.0 kDa	2	0	0		routine	Alms1, Cfap47
362.0-364.0 kDa	1	0	0		routine	Unc80
364.0-366.0 kDa	4	0	0		routine	Aspm, Golgb1, Dchs2, Cdh23
366.0-368.0 kDa	4	0	0	Lamin family	routine	Lama3 [Lamin family], Lrrc37a, Vcan, Vcan
368.0-370.0 kDa	6	0	0	Actin family	routine	Muc4, Ahnak2, Macf1 [Actin family], Cdh23, Cdh23, Cdh23
370.0-372.0 kDa	3	0	0		routine	Golgb1, Unc80, Brca2
372.0-374.0 kDa	1	0	0		routine	Muc5ac
374.0-376.0 kDa	1	0	0		routine	Ttn
376.0-378.0 kDa	2	0	0		routine	Lrp1b, Szt2
380.0-382.0 kDa	1	0	0		routine	Eppk1
386.0-388.0 kDa	4	0	1	NF-kB/Rel	routine	Csmc3, Slep1, Reln [NF-kB/Rel], Csmc1
390.0-392.0 kDa	1	0	0		routine	Gm58671
392.0-394.0 kDa	6	0	0		routine	Zfhx4, Wdfy3, Csmc2, Utrn, Wdfy3, Spata31h-ps1
394.0-396.0 kDa	4	0	0	Actin family	routine	Wdfy3, Zfhx4, Myo15a, Spen [Actin family]
396.0-398.0 kDa	1	0	0	Actin family	routine	Xirp2 [Actin family]
398.0-400.0 kDa	5	0	0	Actin family; Heat shock proteins	routine	Zfhx4, Hspg2 [Heat shock proteins], Spen [Actin family], Spen [Actin family], Cubn
402.0-404.0 kDa	1	0	0		routine	Zfp469
404.0-406.0 kDa	5	0	0	Lamin family	routine	Lama5 [Lamin family], Trrap, Cmya5, Lrrc37, Csmc3
406.0-408.0 kDa	2	0	0		routine	Zfhx3, Zfhx3
408.0-410.0 kDa	3	0	0		routine	Usp34, Smg1, Usp34
412.0-414.0 kDa	3	0	0		routine	Alms1, Mgam, Cmya5
414.0-416.0 kDa	1	0	0		routine	Sptbn5
418.0-420.0 kDa	1	0	0		routine	Bsn
420.0-422.0 kDa	1	0	0		routine	Vps13c
422.0-424.0 kDa	1	0	0		routine	Muc4
424.0-426.0 kDa	3	0	0		routine	Lyst, Lyst, Dmd
426.0-428.0 kDa	2	0	0		routine	Ank2, Ank2
428.0-430.0 kDa	3	0	0	Actin family; Histone family	routine	Xirp2 [Actin family], Ank2, Kmt2a [Histone family]
434.0-436.0 kDa	5	0	0		routine	Trrap, Akap9, Trnb, Trrap, Akap9
436.0-438.0 kDa	2	0	0		routine	Trrap, Trrap

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
442.0-444.0 kDa	1	0	0		routine	Fras1
444.0-446.0 kDa	2	0	0		routine	Vps13b, Pkhd1
446.0-448.0 kDa	1	0	0		routine	Trxb
460.0-462.0 kDa	2	0	0		routine	Dnah7a, Dnah7b
462.0-464.0 kDa	2	0	0		routine	Dnah7c, Muc17
464.0-466.0 kDa	3	0	0		routine	Pkhd1l1, Pkhd1l1, Pkhd1l1
466.0-468.0 kDa	4	0	0		routine	Pkd1, Dnah7b, Dnah3, Dnah3
468.0-470.0 kDa	3	0	0	Heat shock proteins	routine	Dnah6, Hspg2 [Heat shock proteins], Hspg2 [Heat shock proteins]
470.0-472.0 kDa	1	0	1		routine	Prkdc
474.0-476.0 kDa	2	0	0		routine	Trxb, Dnah6
478.0-480.0 kDa	1	0	0		routine	Muc2
480.0-482.0 kDa	1	0	0		routine	Fat2
482.0-484.0 kDa	2	0	0		routine	Huwe1, Huwe1
484.0-486.0 kDa	1	0	0		routine	Hectd4
486.0-488.0 kDa	2	0	0		routine	Dnah1, Vps13d
488.0-490.0 kDa	1	0	0		routine	Gm57895
490.0-492.0 kDa	3	0	0		routine	Gm57894, Vps13d, Unc13b
492.0-494.0 kDa	1	0	0		routine	Dync2h1
498.0-500.0 kDa	1	0	0		routine	Plec
500.0-502.0 kDa	2	0	0		routine	Stard9, Fat3
502.0-504.0 kDa	2	0	0		routine	Fat3, Apob
504.0-506.0 kDa	3	0	0	Actin family	routine	Macf1 [Actin family], Lrp1, Lrp1
506.0-508.0 kDa	2	0	0		routine	Fat1, Fat1
508.0-510.0 kDa	4	0	0		routine	Dnah17, Mycbp2, Apob, Dnah17
510.0-512.0 kDa	2	0	0		routine	Dnah2, Dnah17
512.0-514.0 kDa	5	0	0		routine	Fat1, Dnah2, Dnah9, Lrp1b, Lrp1b
514.0-516.0 kDa	2	0	0		routine	Muc5b, Muc6
516.0-518.0 kDa	2	0	0		routine	Dnah11, Dnah14
518.0-520.0 kDa	1	0	0		routine	Lrp2
520.0-522.0 kDa	3	0	0		routine	Sacs, Sacs, Mycbp2
522.0-524.0 kDa	1	0	0		routine	Dnah10
526.0-528.0 kDa	2	0	0		routine	Herc2, Dnah5
528.0-530.0 kDa	3	0	0		routine	Birc6, Dnah14, Dnah10
530.0-532.0 kDa	1	0	0		routine	Birc6
532.0-534.0 kDa	3	0	0		routine	Dync1h1, Birc6, Herc1
534.0-536.0 kDa	3	0	0		routine	Plec, Gm58669, Sspo
536.0-538.0 kDa	2	0	0		routine	Dnhd1, Dnhd1
540.0-542.0 kDa	4	0	0	Histone family	routine	Kmt2c [Histone family], Kmt2c [Histone family], Fat4, Dnah8
546.0-548.0 kDa	1	0	0		routine	Hmcn2
550.0-552.0 kDa	7	0	0		routine	Sspo, Ryr3, Pclo, Sspo, Ryr3, Spata31h1, Ryr3
554.0-556.0 kDa	2	0	0		routine	Ryr3, Btlp1
564.0-566.0 kDa	2	0	0		routine	Ryr2, Ryr1
568.0-570.0 kDa	4	0	0		routine	Ryr1, Gm57885, Abca13, Ush2A
572.0-574.0 kDa	1	0	0		routine	Ubr4
578.0-580.0 kDa	1	0	0		routine	Zan
580.0-582.0 kDa	1	0	0		routine	Hydin
584.0-586.0 kDa	4	0	0		routine	Rnf213, Rnf213, Zan, Rnf213
590.0-592.0 kDa	1	0	0		routine	Dst
600.0-602.0 kDa	2	0	0	Histone family	routine	Kmt2d [Histone family], Kmt2d [Histone family]
604.0-606.0 kDa	1	0	0		routine	Ahnak
606.0-608.0 kDa	3	0	0	Actin family	routine	Macf1 [Actin family], Macf1 [Actin family], Macf1 [Actin family]
608.0-610.0 kDa	1	0	0	Actin family	routine	Macf1 [Actin family]
610.0-612.0 kDa	1	0	0		routine	Hmcn1
628.0-630.0 kDa	1	0	0		routine	Mdn1
630.0-632.0 kDa	1	0	0		routine	Mdn1
660.0-662.0 kDa	1	0	0		routine	Efcab3
686.0-688.0 kDa	2	0	0		routine	Adgrv1, Adgrv1
692.0-694.0 kDa	1	0	0		routine	Muc19
700.0-702.0 kDa	1	0	0		routine	Gm58672
706.0-708.0 kDa	1	0	0		routine	Fig2
722.0-724.0 kDa	1	0	0		routine	Eppk1
724.0-726.0 kDa	1	0	0		routine	Obscn
746.0-748.0 kDa	1	0	0		routine	Ccdc168
780.0-782.0 kDa	1	0	0		routine	Obscn
782.0-784.0 kDa	2	0	0		routine	Syne2, Syne2
784.0-786.0 kDa	1	0	0	Actin family	routine	Fsip2 [Actin family]
792.0-794.0 kDa	3	0	0		routine	Gm58665, Gm58676, Gm58666
800.0-802.0 kDa	1	0	0		routine	Neb
814.0-816.0 kDa	1	0	0		routine	Obscn
828.0-830.0 kDa	1	0	0		routine	Neb
830.0-832.0 kDa	2	0	0	Actin family	routine	Macf1 [Actin family], Macf1 [Actin family]
832.0-834.0 kDa	1	0	0		routine	Gm58662
834.0-836.0 kDa	1	0	0		routine	Dst
866.0-868.0 kDa	1	0	0		routine	Neb
870.0-872.0 kDa	3	0	0		routine	Dst, Gm61593, Neb
874.0-876.0 kDa	1	0	0		routine	Obscn
908.0-910.0 kDa	1	0	0		routine	Muc16
914.0-916.0 kDa	1	0	0		routine	Ttn
966.0-968.0 kDa	1	0	0		routine	Obscn
1008.0-1010.0 kDa	2	0	0		routine	Syne1, Syne1
2990.0-2992.0 kDa	1	0	0		routine	Ttn
3716.0-3718.0 kDa	1	0	0		routine	Ttn

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
3906.0-3908.0 kDa	1	0	0		routine	Ttn
3924.0-3926.0 kDa	1	0	0		routine	Ttn

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
0.66	Gm45021	A0A1Y7VHP7	Predicted gene 45021 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.72		P38639	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.72	Aars1	A0A1Y7VE86	Alanyl-tRNA synthetase 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.75	Ighd1-1	A0A0G2JDD8	Immunoglobulin heavy diversity 1-1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.76	Tsc2	A0A2I3BRA1	TSC complex subunit 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.77		P38642	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.77	Tlr3	A0A1Y7V141	Toll-like receptor 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.78	Tor1aip2	A0A1Y7VDJ1	Torsin A interacting protein 2 (Fragment)	0.0-2.0 kDa	Actin family	configured audit family match: Actin family
0.78	Dtx2	A0A1C7ZM26	Deltex 2, E3 ubiquitin ligase (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.78	Slc2a4	A0A0G2JDD0	Solute carrier family 2 (facilitated glucose transporter), member 4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.79	Pou2f1	A0A1Y7VFH6	POU domain, class 2, transcription factor 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.79	Ighd1-3	A0A1Y7VHD7	Predicted gene, 16968 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.80	Rnf44	A0A1Y7VGK4	Ring finger protein 44 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.81	Dab2	A0A1Y7VGS2	Disabled 2, mitogen-responsive phosphoprotein (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.81	Pgpep1	A0A1B0GT82	Pyroglutamyl-peptidase I (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.82	Nrip1	A0A1Y7VG73	Nuclear receptor interacting protein 1 (Fragment)	0.0-2.0 kDa	Actin family	configured audit family match: Actin family
0.82	Tll1	A0A1B0GRW8	Tolloid-like (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.82		P38644	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.83	Zbtb18	A0A1Y7VJC4	Zinc finger and BTB domain containing 18 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.83	Plgrkt	A0A1Y7VE20	Plasminogen receptor, C-terminal lysine transmembrane protein (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.83	Slc16a3	A0A0G2JD98	Solute carrier family 16 (monocarboxylic acid transporters), member 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.84		P38641	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.86	Pcdh19	A0A1Y7VLJ9	Protocadherin 19 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.88	Actf6b	A0A1C7ZN05	Actin-like 6B (Fragment)	0.0-2.0 kDa	Actin family	configured audit family match: Actin family
0.89		P99031	Unknown protein from spot 2D-0014LD of 2D-PAGE of liver tissue (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.89	St3gal6	A0A338P6H0	ST3 beta-galactoside alpha-2,3-sialyltransferase 6 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.90	Itprlp	A0A494B9D9	Inositol 1,4,5-triphosphate receptor interacting protein (Fragment)	0.0-2.0 kDa	Actin family	configured audit family match: Actin family
0.90	Sulf1	A0A0G2JDD4	Sulfatase 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.90	Dedd	A0A1C7ZM28	Death effector domain-containing (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.91	Mis12	A0A0G2JDA2	MIS12 kinetochore complex component (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.91	Tgtp1	A0A1L1STI2	T cell specific GTPase 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.92	Tmc6	A0A0G2JDA0	Transmembrane channel-like gene family 6 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.93	Cln8	A0A1C7ZM20	CLN8 transmembrane ER and ERGIC protein (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.93	Plkp	A0A0G2JD99	Phosphofructokinase, platelet (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.93	Sv2b	A0A0U1RQ76	Synaptic vesicle glycoprotein 2b (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.95	Socs6	A0A1C7ZMY9	Suppressor of cytokine signaling 6 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.95	Cox6c	A0A0G2JDA6	Cytochrome c oxidase subunit 6C (Fragment)	0.0-2.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
0.95	Ccdc9	A0A0G2JDC5	Coiled-coil domain containing 9 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.95	Cdk17	A0A1L1SSL3	Cyclin dependent kinase 17 (Fragment)	0.0-2.0 kDa	Cell-cycle CDK/cyclin	configured signaling gene; check phospho/total controls and antibody validation; configured audit family match: Cell-cycle CDK/cyclin
0.96	Snap47	A0A0G2JDA4	Synaptosomal-associated protein, 47 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.97	Ighd6-4	A0A0G2JDD5	Immunoglobulin heavy diversity 5-7 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.97	Slc4a2	A0A1C7ZN01	Solute carrier family 4 (anion exchanger), member 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.98	P2ry14	A0A0G2JGB4	Purinergic receptor P2Y, G-protein coupled, 14 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.98	Dhx32	A0A1C7ZM23	DEAH (Asp-Glu-Ala-His) box polypeptide 32 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.98	Plekhh3	A0A1W2P798	Pleckstrin homology domain containing, family G (with RhoGef domain) member 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.99		P38640	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.00	Tmod2	A0A1L1ST54	Tropomodulin 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.01	Gfra1	A0A1C7ZN00	Glial cell line derived neurotrophic factor family receptor alpha 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.02	Fbxo8	A0A1B0GSY6	F-box protein 8 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.02	Sema3b	A0A0G2JDD7	Sema domain, immunoglobulin domain (Ig), short basic domain, secreted, (semaphorin) 3B (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.02	Cyth3	A0A0G2JDC7	Cytohesin 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.03	Dnajb4	A0A0G2JEM9	DnaJ heat shock protein family (Hsp40) member B4 (Fragment)	0.0-2.0 kDa	Heat shock proteins	configured audit family match: Heat shock proteins
1.05	Cited2	A0A1W2P7B1	Cbp/p300-interacting transactivator, with Glu/Asp-rich carboxy-terminal domain, 2 (Fragment)	0.0-2.0 kDa	Actin family	configured audit family match: Actin family
1.06	Phf21a	A0A1C7ZN07	PHD finger protein 21A (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.07	Atosb	A0A0G2JDA3	Atos homolog B (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.07	Ifi47	A0A1L1SSW0	Interferon gamma inducible protein 47 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.07	Tpm3	A0A1C7ZMY7	Tropomyosin 3, gamma (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.07	Arhgap39	H3BJW7	Rho GTPase activating protein 39 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.08		P83853	Vam6-like protein (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.08	Zfp62	A0A0G2JDA7	Zinc finger protein 62 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.09	Ccnl1	F7BER3	Cyclin L1 (Fragment)	0.0-2.0 kDa	Cell-cycle CDK/cyclin	configured signaling gene; check phospho/total controls and antibody validation; configured audit family match: Cell-cycle CDK/cyclin
1.09	Srsf3	A0A3B2W3M9	Serine and arginine-rich splicing factor 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.09	Cbarp	D3YXP8	Calcium channel, voltage-dependent, beta subunit associated regulatory (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.10	Hhatf	A0A1L1SUF7	Hedgehog acyltransferase-like (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.10		P99032	Unknown protein from spot 2D-0014M9 of 2D-PAGE of liver tissue (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.11		P38643	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.11	Srd5a3	A0A1C7ZMY6	Steroid 5 alpha-reductase 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.11	Lsp1	A0A1B0GSH0	Lymphocyte specific 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.11	Oxnad1	E9Q7K0	Oxidoreductase NAD-binding domain containing 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.11	Snd1	A0A1C7ZN09	Staphylococcal nuclease and tudor domain containing 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.11	Mtf1	A0A0G2JDC6	Metal response element binding transcription factor 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
1.11	March8	A0A0N4SVW5	Membrane associated ring-CH-type finger 8 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.12	Rsrc2	A0A0G2JDD3	Arginine/serine-rich coiled-coil 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.12	Huwe1	A0A0G2JD97	HECT, UBA and WWE domain containing 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.12	Henmt1	A0A0G2JFU0	HEN1 methyltransferase homolog 1 (Arabidopsis) (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.13	Fbln2	A0A1C7ZM29	Fibulin 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.14	Snph	A0A140T8V7	Syntaphilin (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.14	Zfp266	A0A1L1SU03	Zinc finger protein 266 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.15	Slc7a2	A0A1C7ZMY5	Solute carrier family 7 (cationic amino acid transporter, y+ system), member 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.15	Vangl1	A0A140T8S8	VANGL planar cell polarity 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.15	Nhs13	D3Z770	NHS like 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.16	Zfp219	A0A213BQ96	Zinc finger protein 219 (Fragment)	0.0-2.0 kDa	p53 pathway	configured audit family match: p53 pathway
1.16	Apb2	E0CYB2	Amyloid beta precursor protein binding family B member 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.17	Zfp40	B1B1D4	Zinc finger protein 40 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.17	Nrbp2	A0A213BRU8	Nuclear receptor binding protein 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.17	Igjl3p	A0A0G2JFS3	Immunoglobulin lambda joining 3 pseudogene (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.17	Ltf	A0A0G2JDK4	Lactotransferrin (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.18	Phf21a	A0A1C7ZN08	PHD finger protein 21A (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.18	Kctd9	A0A1C7ZM21	Potassium channel tetramerisation domain containing 9 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.19	Zfp57	G3UX77	Zinc finger protein 57 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.19		P83854	Putative G protein-coupled receptor (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.19	Rreb1	B83JE5	Ras responsive element binding protein 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.19	Otoa	E9PWK6	Otoancorin (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.20	Sncap	A0A0G2JDC8	Synuclein, alpha interacting protein (synphilin) (Fragment)	0.0-2.0 kDa	Actin family	configured audit family match: Actin family
1.20	Arap1	D3YV58	ArfGAP with RhoGAP domain, ankyrin repeat and PH domain 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.20	Cnpy1	A0A1C7ZMY8	Canopy FGF signaling regulator 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.20	Vmac	A0A3B2WDC5	Vimentin-type intermediate filament associated coiled-coil protein (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.20	Itgb7	A0A2R8V138	Integrin beta 7 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.20	Lrrc14	A0A1C7ZN04	Leucine rich repeat containing 14 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.22	Slc20a2	A0A1B0GRL3	Solute carrier family 20, member 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.23	Dmac2l	A0A1Y7VMG2	Distal membrane arm assembly component 2 like (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.24	Gata4	F6Y7Q6	GATA binding protein 4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.24	Irx2	F6ZTI9	Iroquois homeobox 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.24	Igjl2	A0A0G2JDZ1	Immunoglobulin lambda joining 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.24	Tceal8	A2AJR9	Transcription elongation factor A (SII)-like 8 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.24	Rec114	A0A087WP95	REC114 meiotic recombination protein (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.24	Gm49385	A0A1L1SR13	Predicted gene, 49385 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.25		A0ABJ3HP72	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.26	Resf1	A0A140T8W0	Retroelement silencing factor 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.27	Tmtc3	D3YWM5	Transmembrane and tetratricopeptide repeat containing 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.27	Igjl3	A0A0G2JDG5	Immunoglobulin lambda joining 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.27	Arhgef28	A0A286YD71	Rho guanine nucleotide exchange factor 28 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.27	Gcc1	E9Q6U7	Golgi coiled coil 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.27	Mmp14	A0A0G2JF39	Matrix metalloproteinase 14 (membrane-inserted) (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.28	Gm42417	A0A140LHU5	Predicted gene, 42417 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.28	Mmg12	A0A140T8T6	Membrane magnesium transporter 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.28	Grsf1	D3YZS7	G-rich RNA sequence binding factor 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.28	Igkj5	A0A0G2JDI1	Immunoglobulin kappa joining 5 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.28	Igjl1	A0A0G2JE61	Immunoglobulin lambda joining 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.28	Ankrd24	A0A1C7ZN03	Ankyrin repeat domain 24 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
ORI_Eckert_JBC2014_MEK3	MEK3 western blot	38.0-40.0 kDa	ORI reported MEK3 bands were compiled from a source image representing p44 expression in an unrelated experiment.	A public ORI case documents a human-cell western blot malpractice example in the 38-40 kDa bin, where MEK3-labeled bands were compiled from an unrelated p44 source image.
ORI_Eckert_JBC2014_p38d	p38 δ western blot	40.0-42.0 kDa	ORI reported p38 δ bands were compiled from a source image representing β -actin expression in an unrelated experiment.	The 40-42 kDa bin is supported by a public ORI case in which p38 δ -labeled bands were compiled from a β -actin source image.
ORI_Eckert_PLoS2012_TAM67	TAM67-FLAG western blot	48.0-50.0 kDa	ORI reported TAM67-FLAG bands were generated using unrelated bands from a source image representing Cyclin A expression.	A public ORI case maps a relabeled western blot problem to the 48-50 kDa neighborhood, involving Cyclin A source material used for a TAM67-FLAG claim.
ORI_Yang_GAPDH_NFYA	GAPDH loading-control bands	36.0-38.0 kDa	ORI reported claimed GAPDH loading-control bands were actually a prominent nonspecific/background band from an NF-YA western blot.	The 36-38 kDa neighborhood is supported by an ORI-described case where a nonspecific NF-YA blot band was claimed as GAPDH.
CancerGeneTherapy_Liu2021_CTBP2_GAPDH	CTBP2 and GAPDH western blots	48.0-50.0 kDa	Retraction note reported apparent similarity in western blots presented for different proteins, including CTBP2 and GAPDH.	A retraction notice maps an apparent different-protein western blot similarity to the CTBP2/GAPDH region, spanning roughly the 36-38 and 48-50 kDa neighborhoods.
CancerGeneTherapy_Liu2021_CyclinD1_CDK4	Cyclin D1 and CDK4 western blots	34.0-36.0 kDa	Retraction note reported apparent similarity in western blots presented for Cyclin D1 and CDK4.	A retraction notice maps an apparent different-protein western blot similarity to the 32-36 kDa cell-cycle neighborhood involving Cyclin D1 and CDK4.
Das_UConn_Garlic_AKT	Akt / p-Akt western blots in Figure 6	54.0-56.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 54-56 kDa signaling region is supported as an audit-priority bin by the garlic-paper retraction, where Akt/p-Akt figure panels were within figures reported to contain fabricated data.
Das_UConn_Garlic_FOXO1	FoxO1 / p-FoxO1 western blots in Figure 6	70.0-72.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 70 kDa transcription-factor signaling region can be referenced as an audit-priority bin because FoxO1 panels occurred in a figure reported to contain fabricated data.
Das_UConn_Garlic_NFKB_PPAR_Glut4	Nrf2, NF-kB p65, PPAR α / δ , Glut-4 western blots in Figures 7 and 8	52.0-54.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 50-56 kDa metabolic/signaling region is an audit-priority region in the garlic-paper case because PPAR and Glut-4 panels were in figures reported to contain fabricated data.
Ilangovan_HSP25_HSP27	Hsp25/Hsp27-family western blots	22.0-24.0 kDa	Public investigation materials described Hsp-family western blot relabeling and unreliable/falsified blot images.	The 22-24 kDa small heat-shock-protein neighborhood can be cited as an audit-risk bin based on public investigation materials describing Hsp-family western blot relabeling.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	mouse_UP000000589.fasta
Marker file	mouse_marker_genes.tsv
Signaling-prefix file	mouse_signaling_prefixes.tsv
Family-pattern file	mouse_family_patterns.tsv
Species/report label	Mus musculus